

AWS DataSync – Copy Data from On-Premises SMB Storage to Amazon S3

To Begin with the Lab

Summary of the Lab

In this lab, **AWS DataSync** was used to copy data from an on-premises-style SMB storage setup into **Amazon S3**. To simulate the on-premises environment, a Windows Server 2012 virtual machine was created in **VMware Workstation**, configured with a shared folder, and given test files to transfer. A DataSync agent was downloaded as an OVA, imported into VMware, assigned a static IP, and activated through the AWS console. After that, a DataSync task was created with the SMB share as the source and an S3 bucket as the destination, using an automatically generated IAM role. The task was started manually, and the execution history confirmed that multiple files were transferred successfully into the S3 bucket.

Understanding AWS DataSync

AWS DataSync is an online data transfer service that simplifies and accelerates moving data between on-premises storage systems and AWS storage services such as **Amazon S3**, **Amazon EFS**, and **Amazon FSx**. It exists so you can migrate, archive, protect, or move data without building a custom copy process. In this lab, the source was an SMB share hosted on a Windows Server virtual machine, and the destination was an S3 bucket in AWS. The benefit is that DataSync handles the transfer securely and automatically, while you focus only on the source and destination locations.

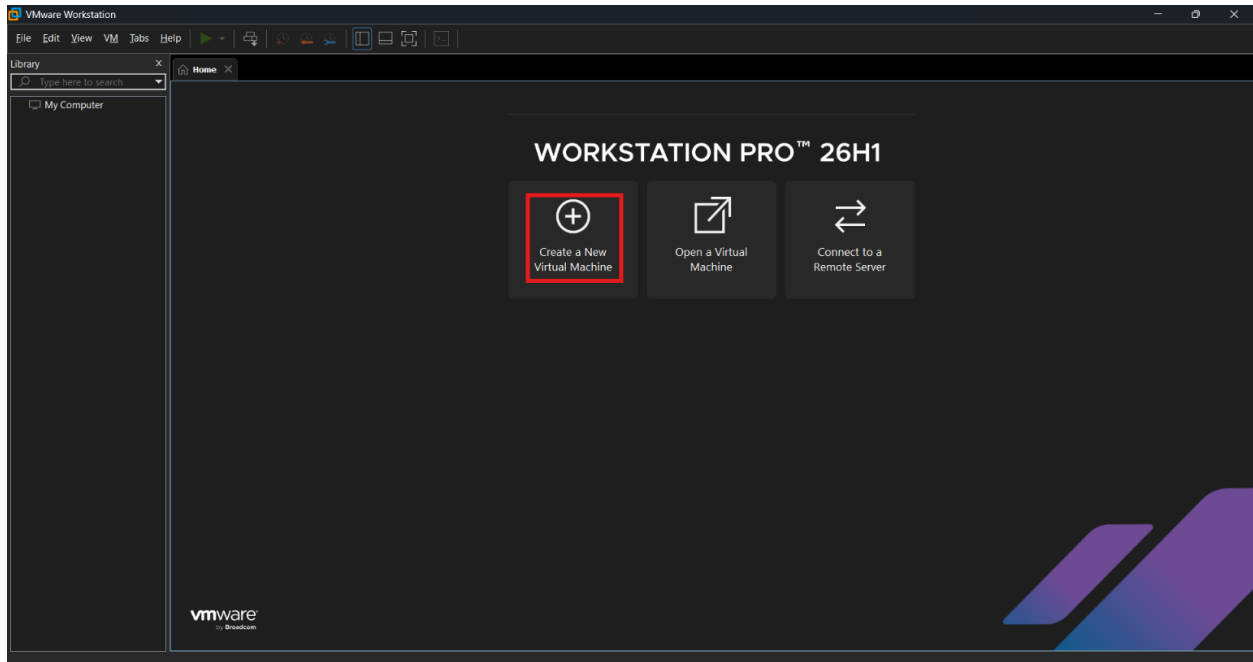
Example:

A company keeps files in a Windows SMB share named data inside a VMware-based storage server. The team wants those files to appear in an S3 bucket named on prem data for backup or processing. DataSync reads the shared folder from the source machine and copies the files into the bucket, so the same files are available in AWS without manual uploading.

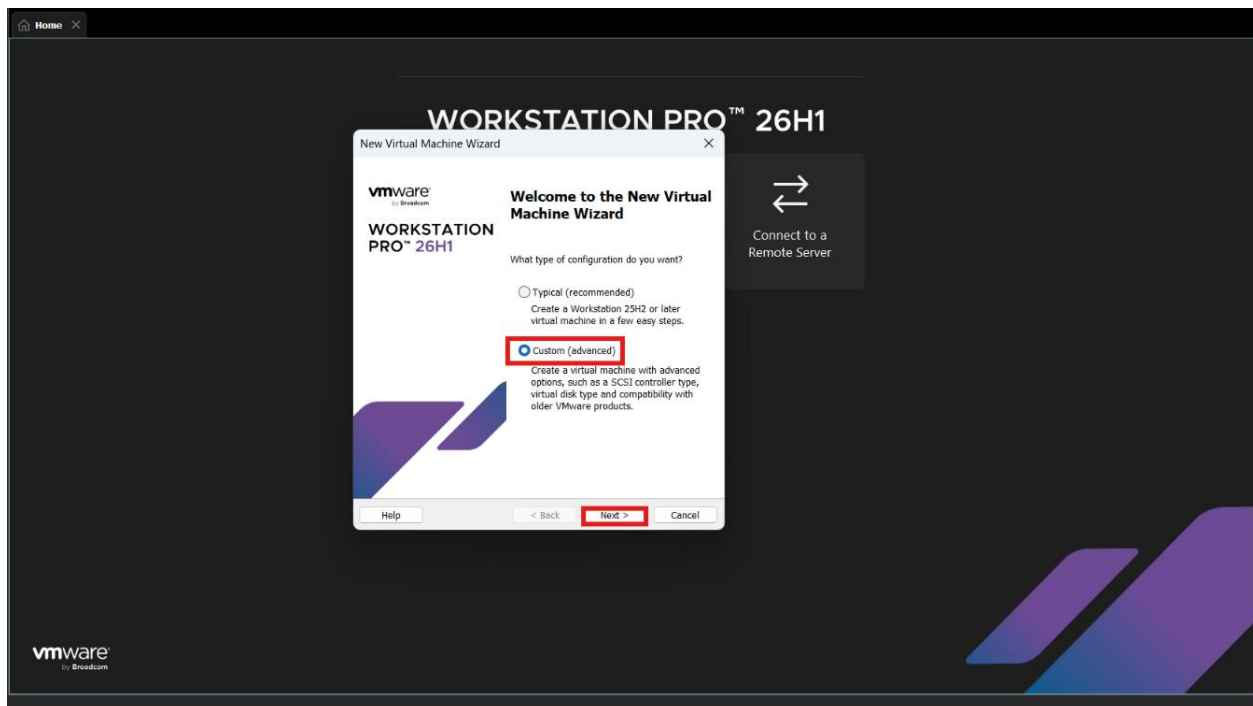
Lab Steps

Step 1 – Prepare the on-premises-style storage server

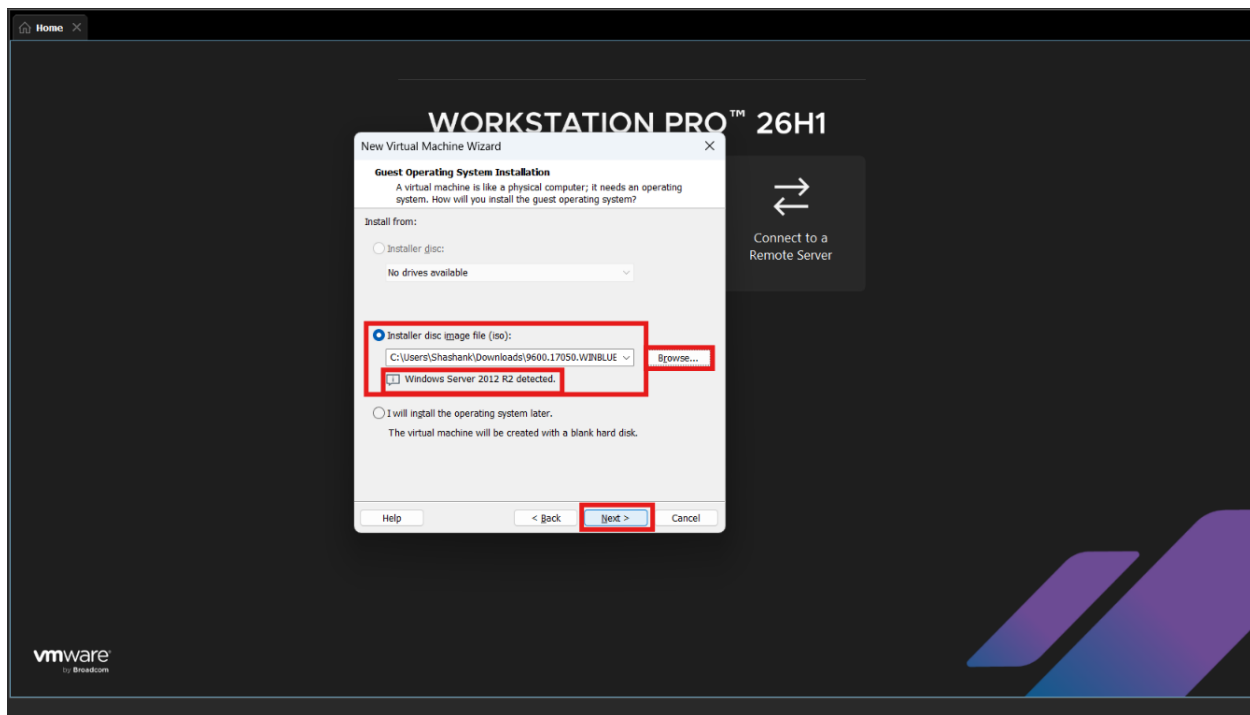
- Open **VMware Workstation**.
- Click on Create New Virtual Machine.



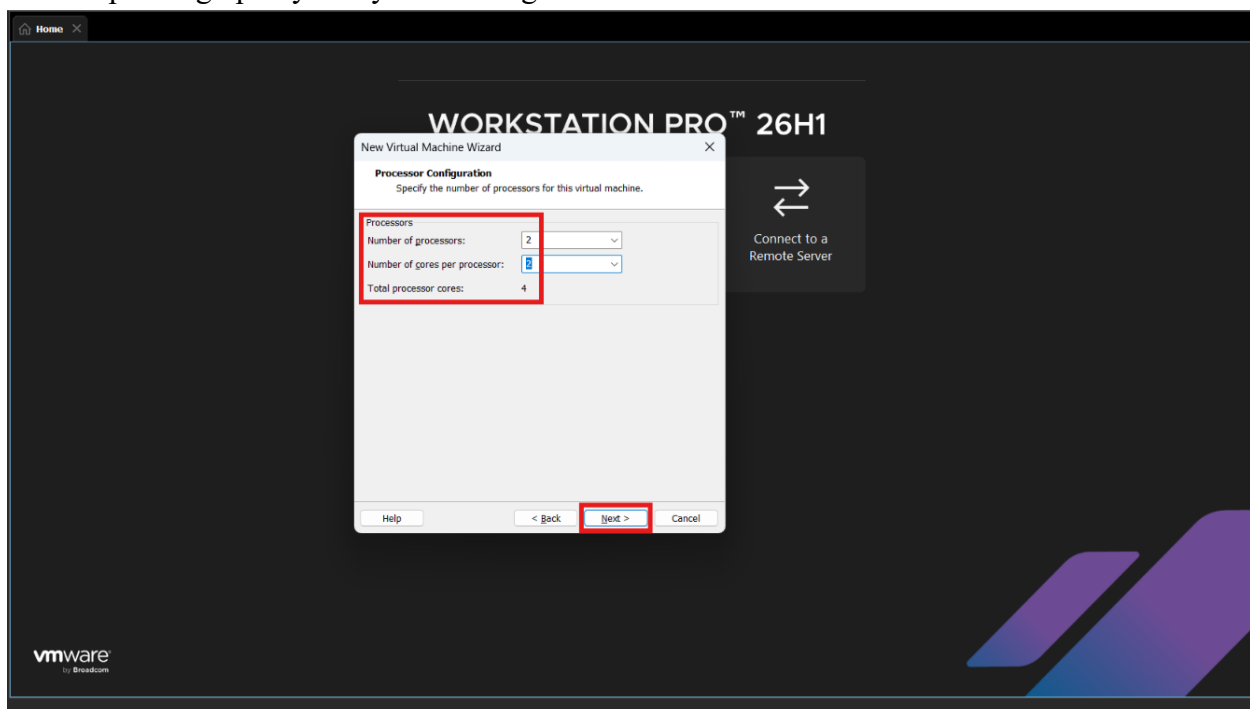
- Create a new virtual machine using **Windows Server 2012** ISO.
- If you don't have the Windows ISO you can download it through its official site.
- For VMWare you have to make an account on their official site to download VMWare Workstation Pro.
- Now choose **Custom** and Click **Next**.



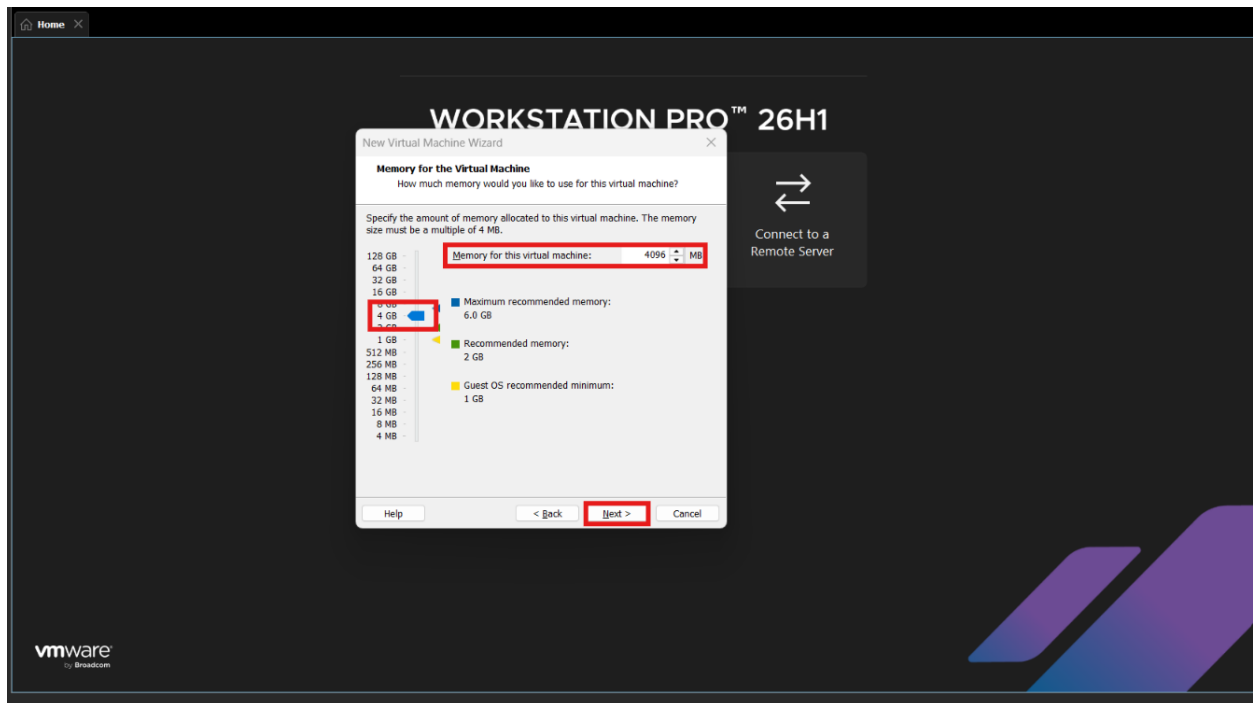
- Choose **Windows 2012 ISO** we have downloaded earlier by clicking on **Browse** and click on **Next**.



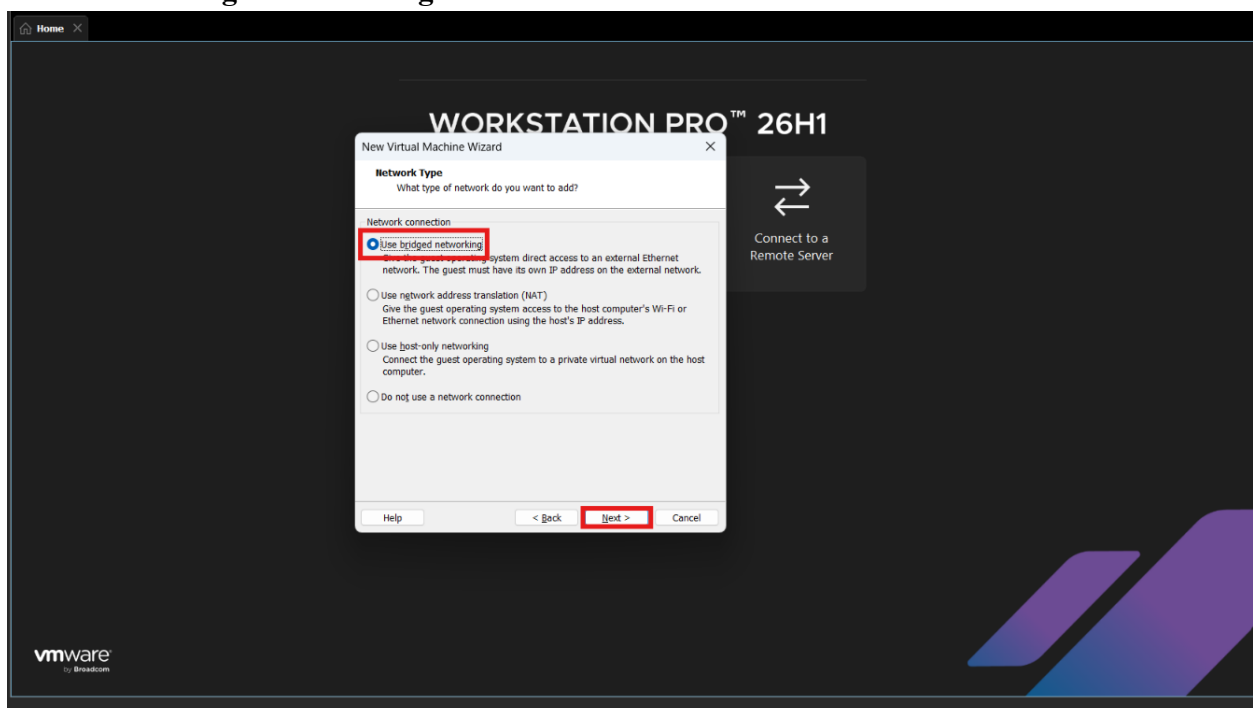
- Now I will assign **4 CPU** based on my CPU Configuration, but you can choose minimum of 2 depending upon your system configurations.



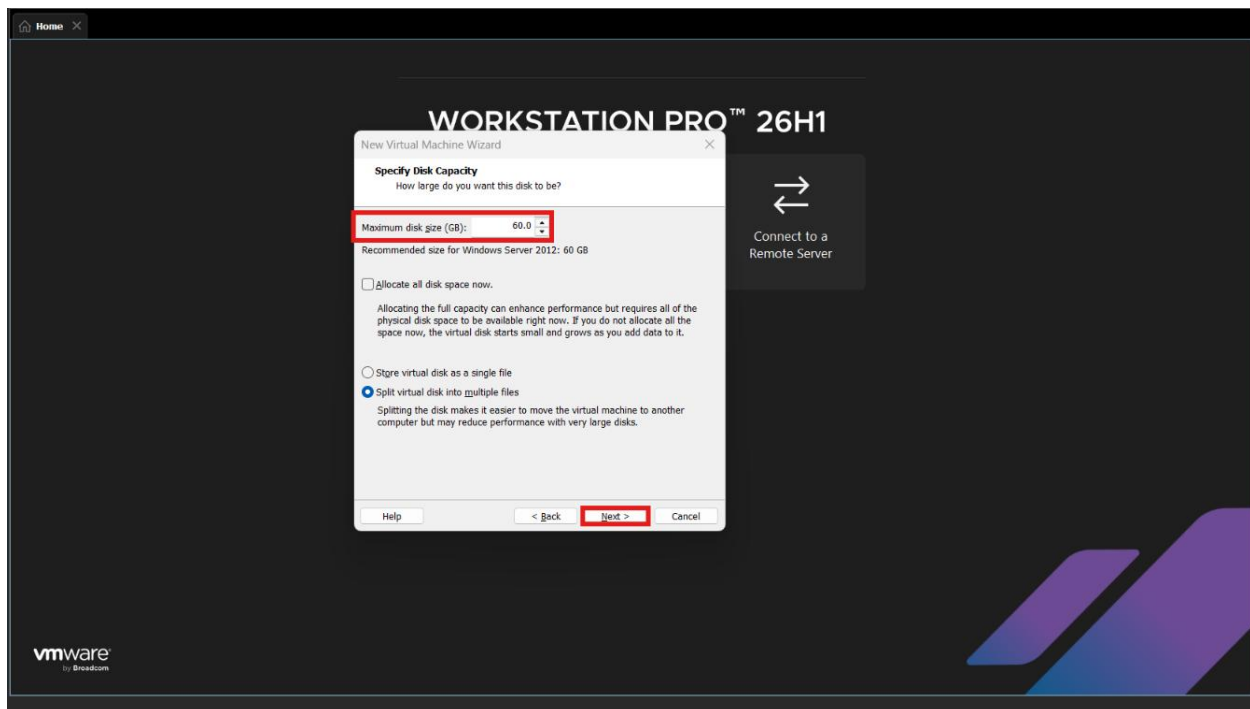
- Now provide at least **4 GB** of RAM for the machine.



- Choose **Bridged networking** and click **Next**.



- Leave rest options as default and keep clicking on **Next**.
- Choose **Maximum disk size** as **60GB** depending on your pc you can configure more as well.



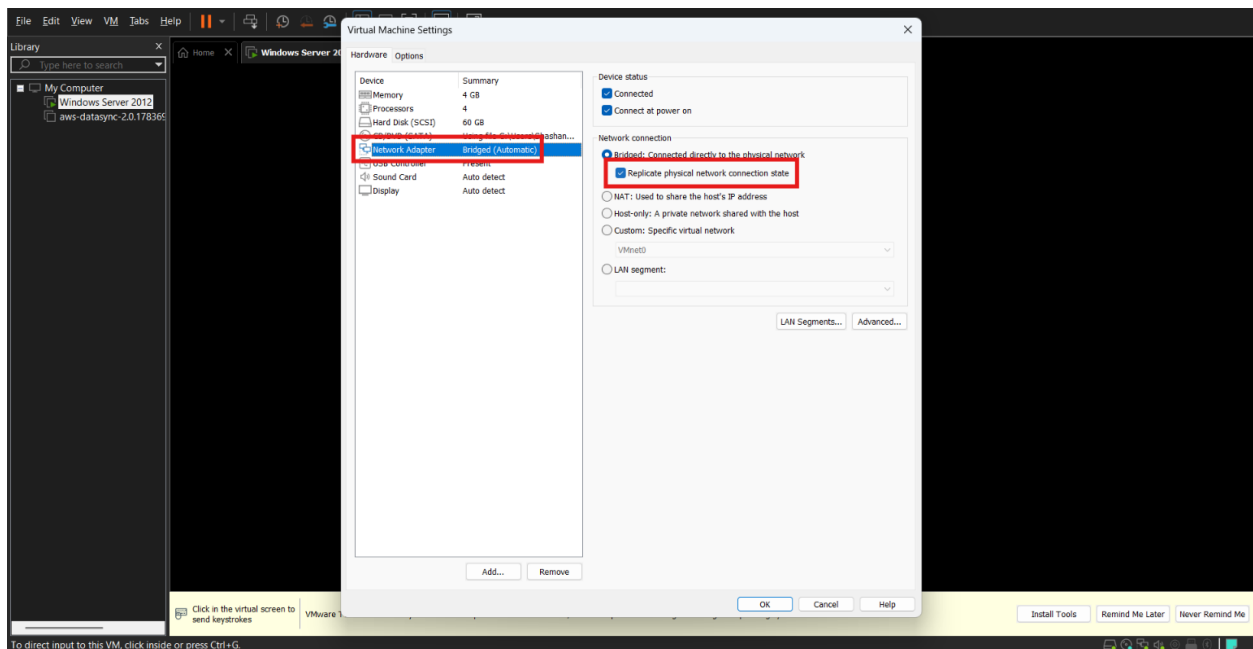
- Finish the VM creation.

Step 2 – Install and configure Windows Server

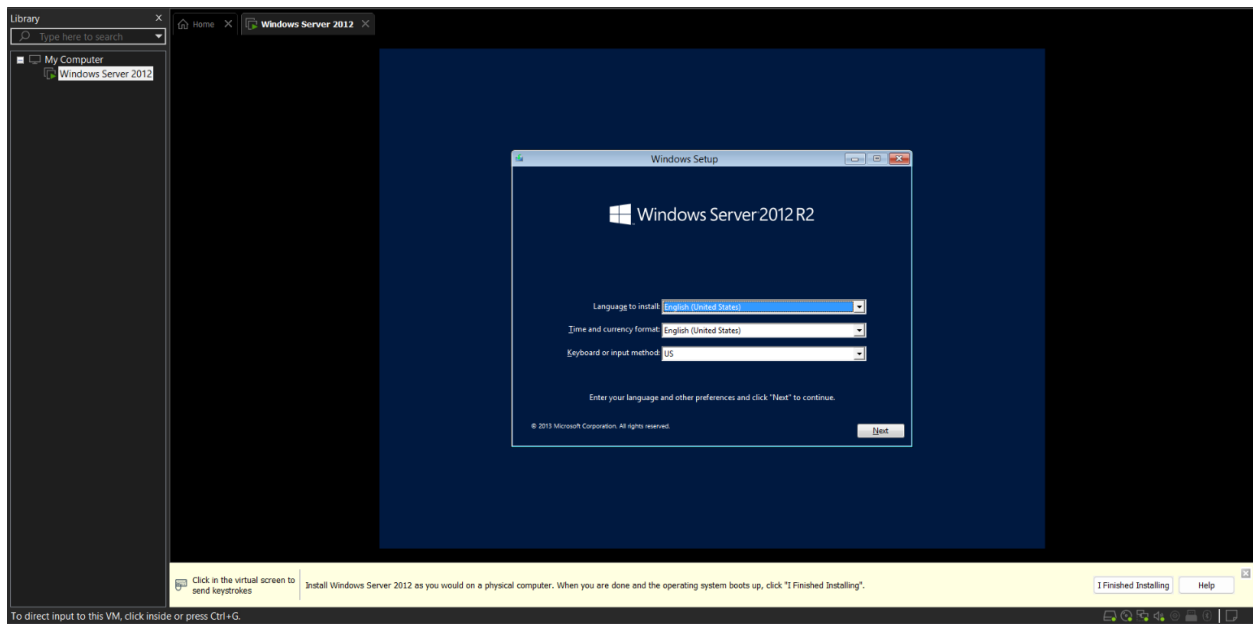
- Right-click on **Virtual Machine** and click on **Settings**.



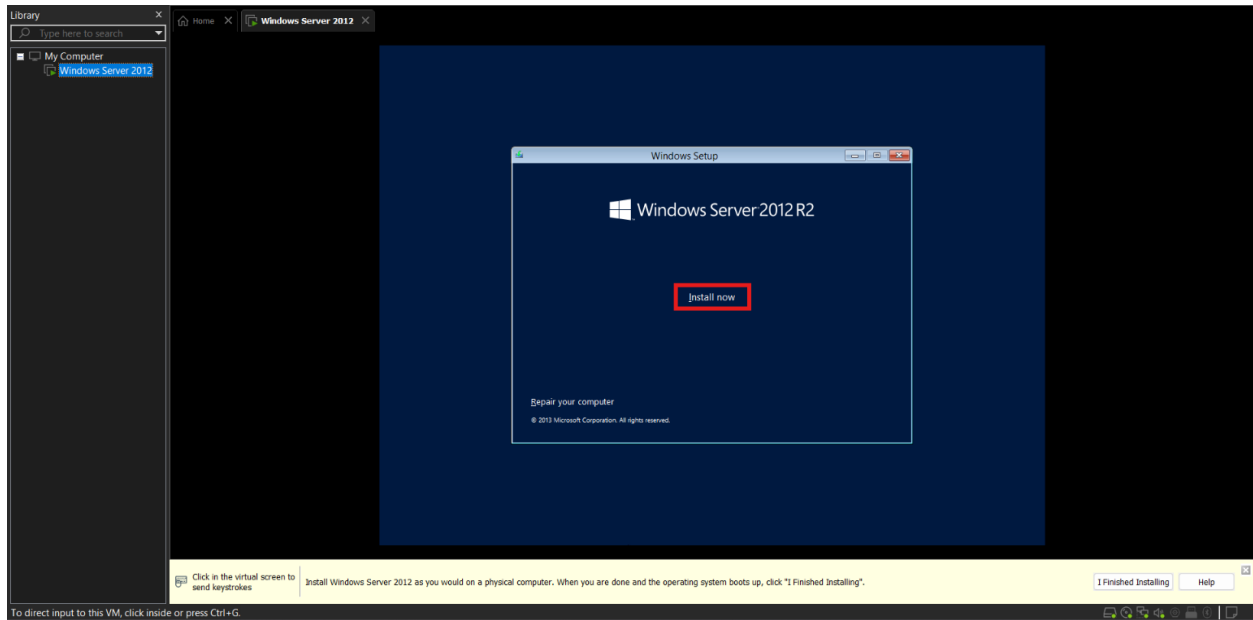
- Now Click on **Network Adapter** and Enable **Replicate physical network state connection**.



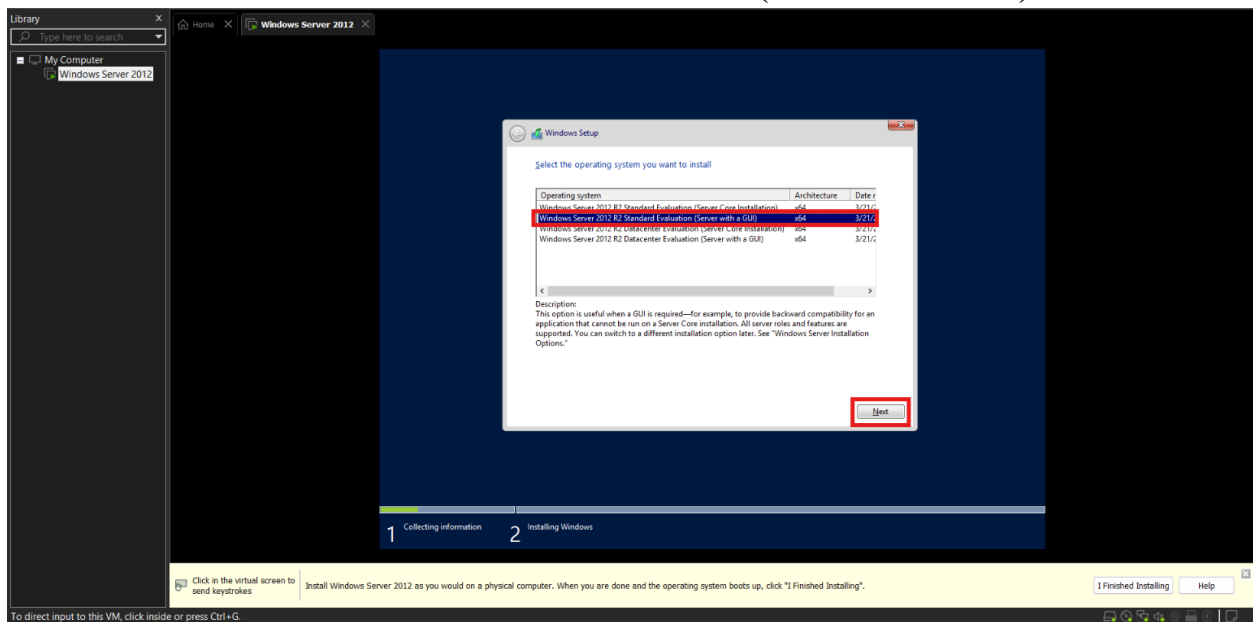
- Power on the virtual machine.
- Install **Windows Server 2012 Standard with GUI**.



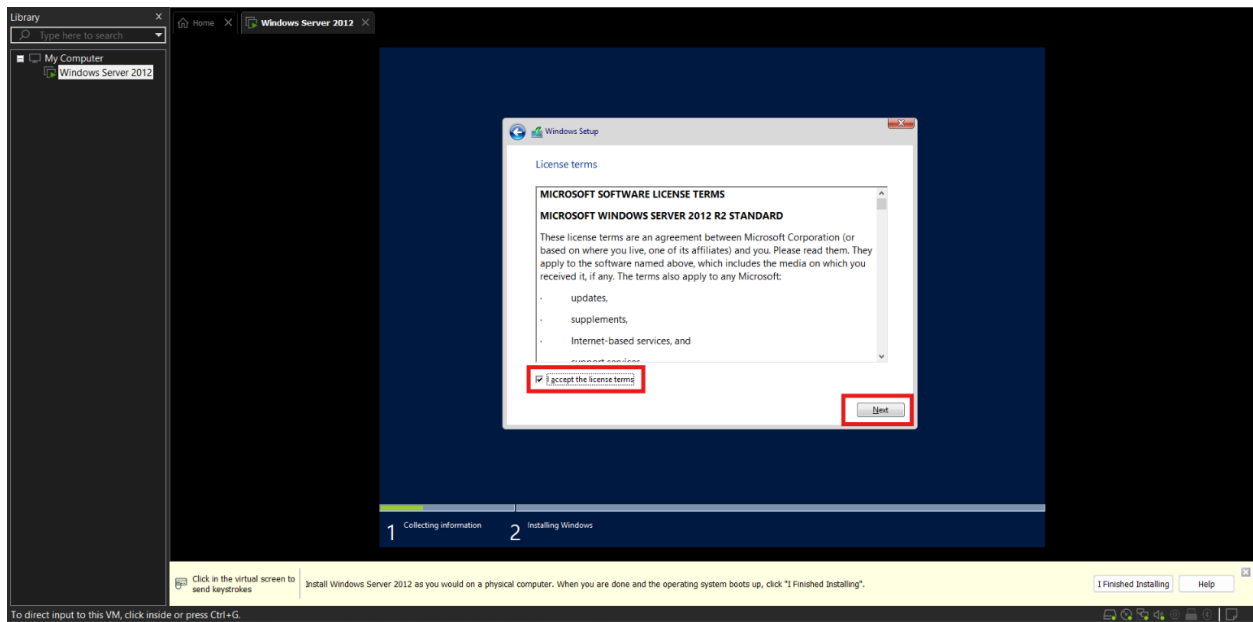
- Click **Next** and choose the **Install Now**.



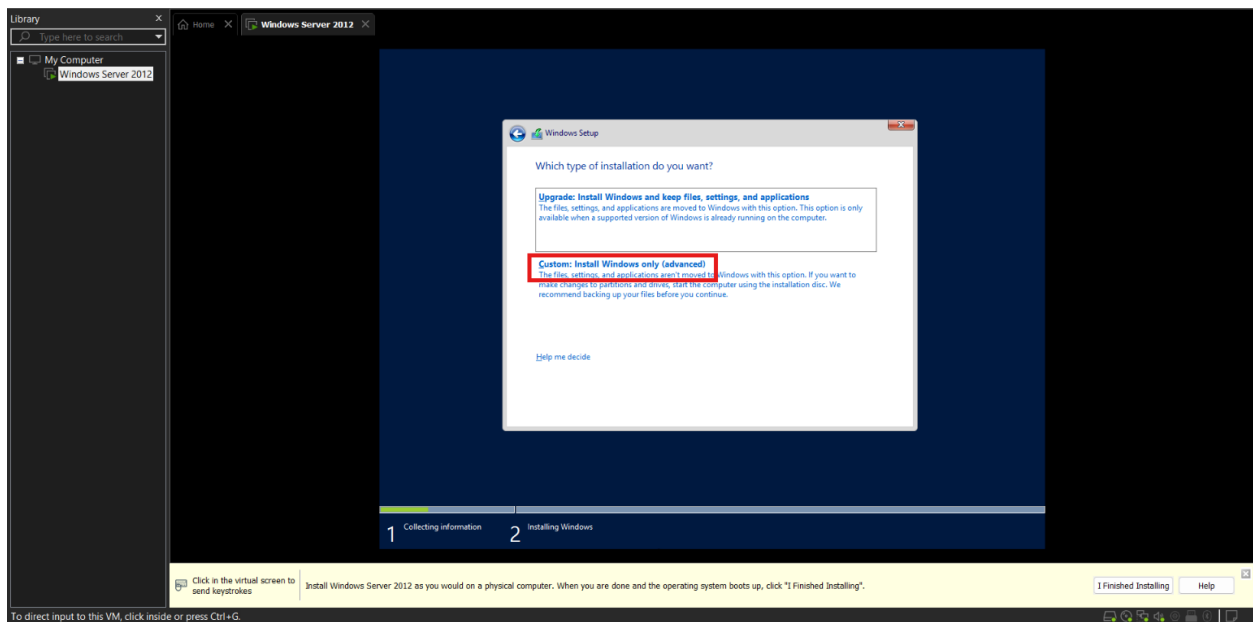
- Choose **Windows Server 2012 Standard Evaluation (Server with a GUI)** and click **Next**.



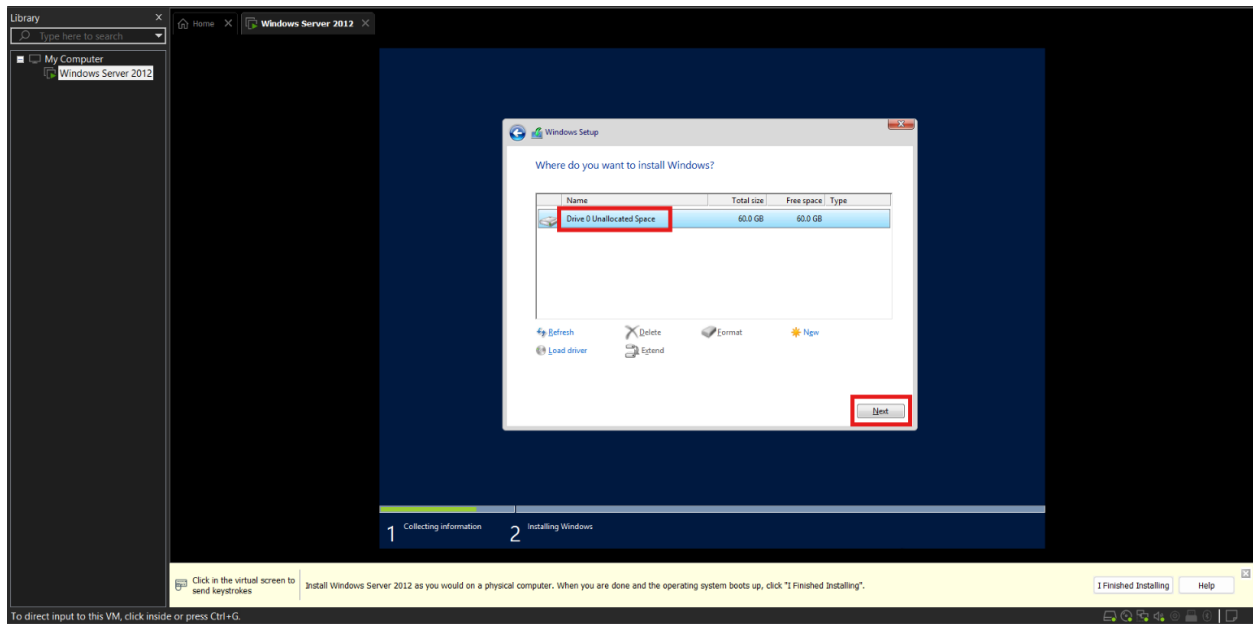
- **Accept License** and click **Next**.



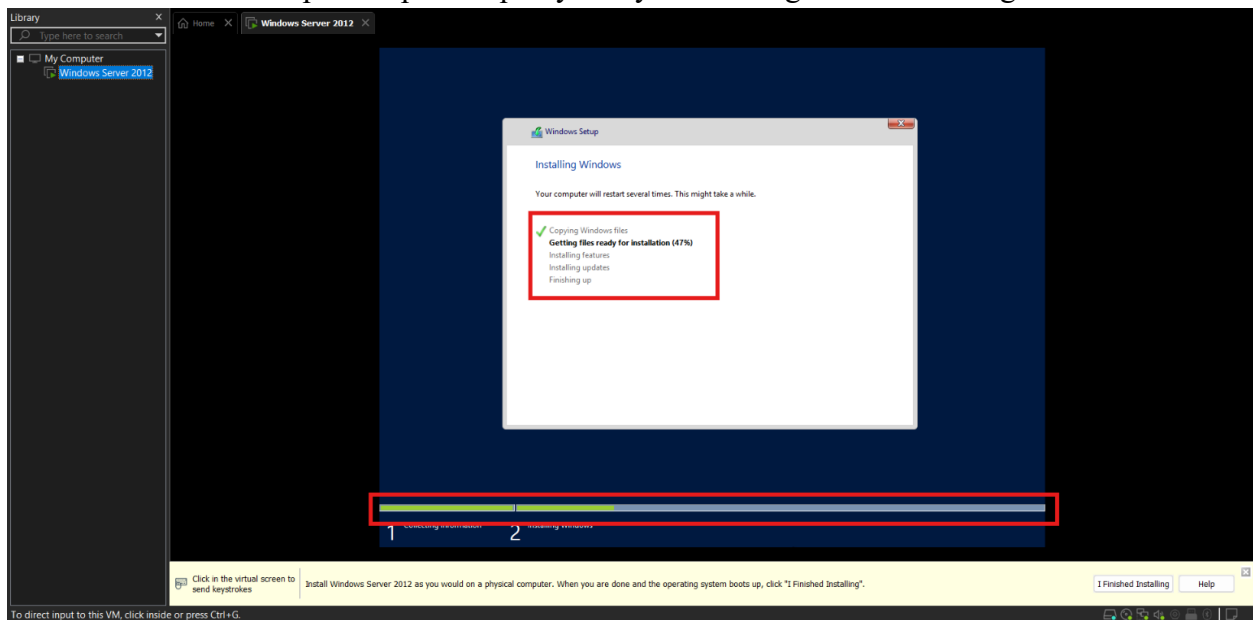
- Click on **Custom**.



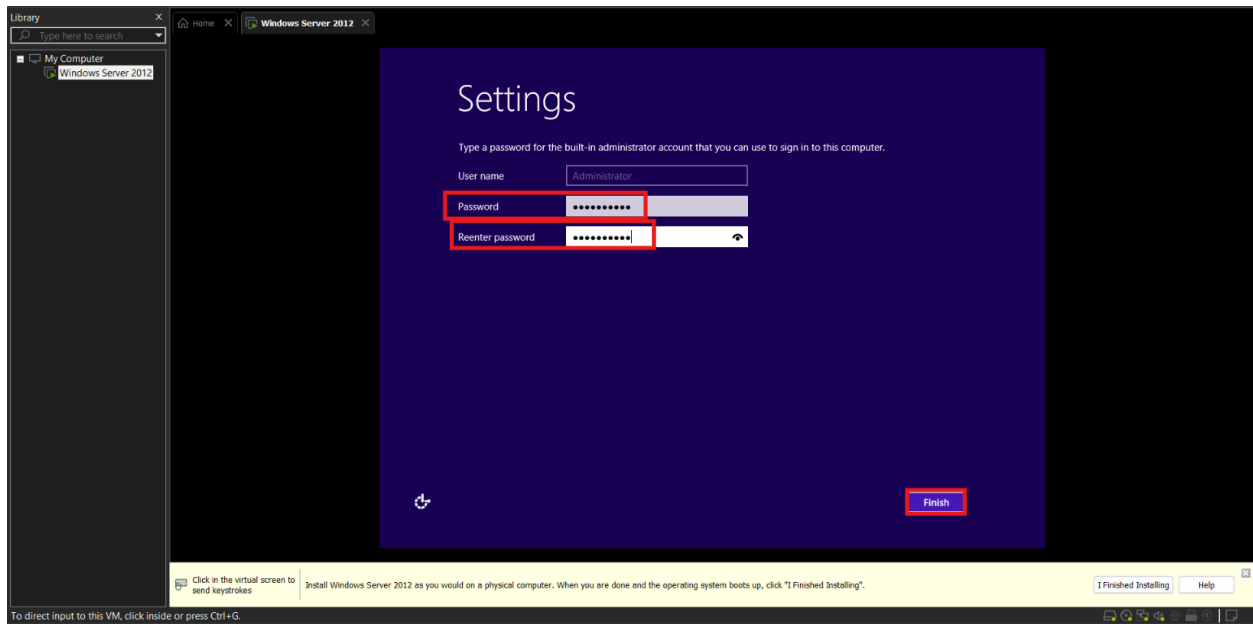
- Now click on the **Drive** and then proceed with installation by clicking on **Next**.



- Let it install as its speed depends upon your system configuration for being installed.

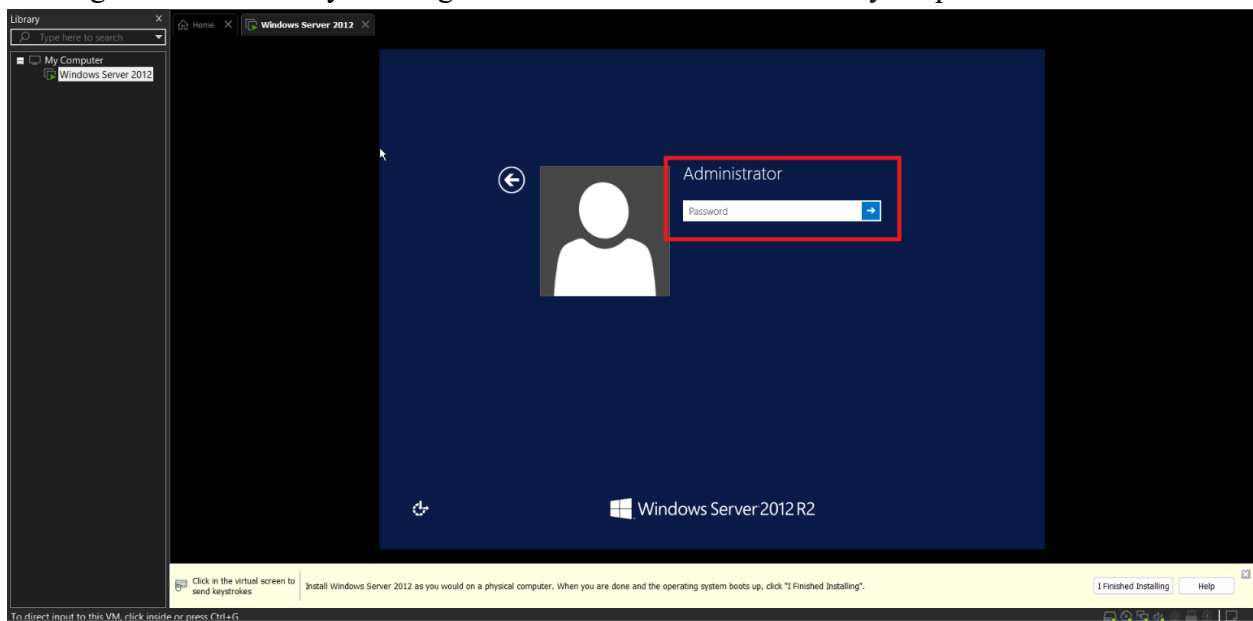


- Set the administrator password and click **Finish**.



Step 3 – Configure IP Settings

- Login to the Server by clicking Ctrl+Alt+Delete and then enter your password.



- Now open **Command Prompt** on your main computer and type **ipconfig** to get your IP Address.

```
Command Prompt
C:\Users\Shashank>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : 
    IPv4 Address. . . . . : 
    Subnet Mask . . . . . : 
    Default Gateway . . . . . :

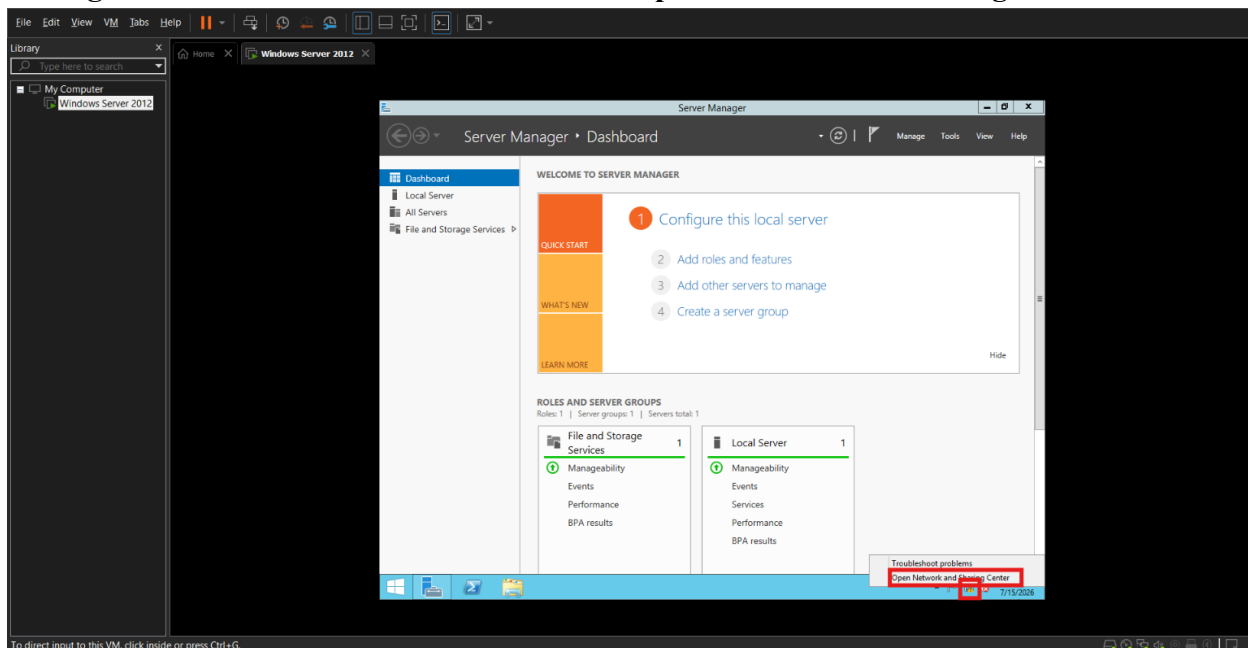
Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : 
    IPv4 Address. . . . . : 
    Subnet Mask . . . . . : 
    Default Gateway . . . . . :

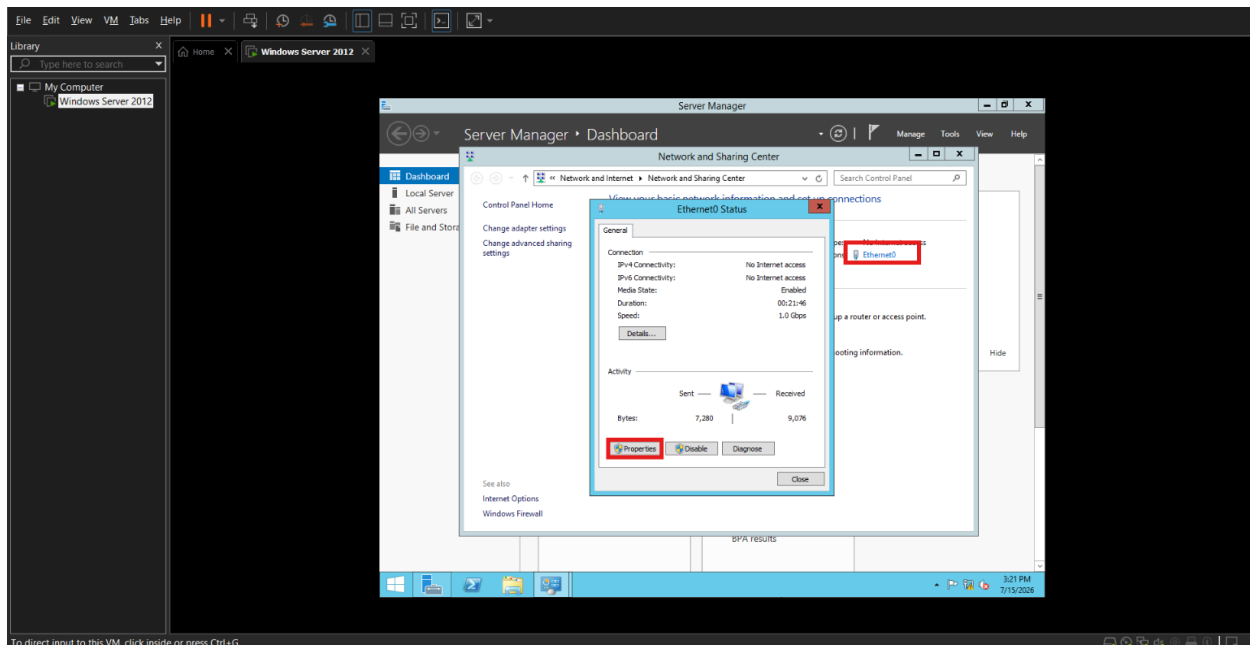
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 
    Temporary IPv6 Address. . . . . : 
    Link-local IPv6 Address . . . . . : 
    IPv4 Address. . . . . : 192.168.1.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

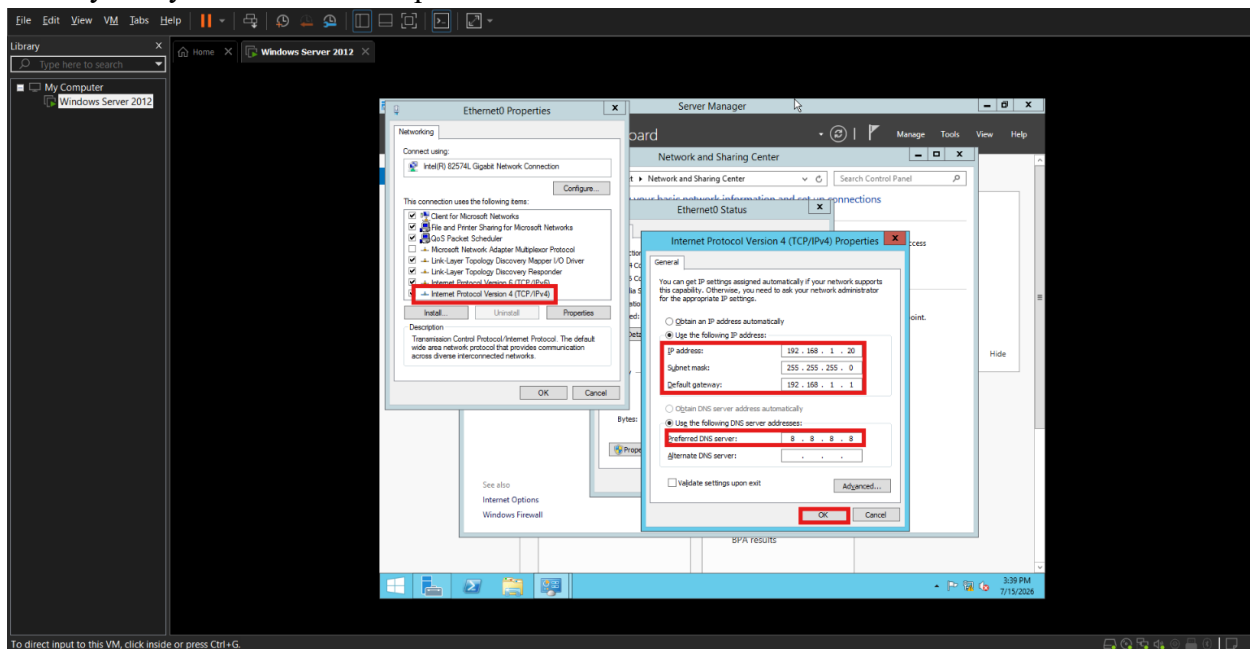
- Now we will configure the same series IP Address on the Virtual Machine so for that we will **Right-Click on Network Icon** and Click on **Open Network and Sharing Center**.



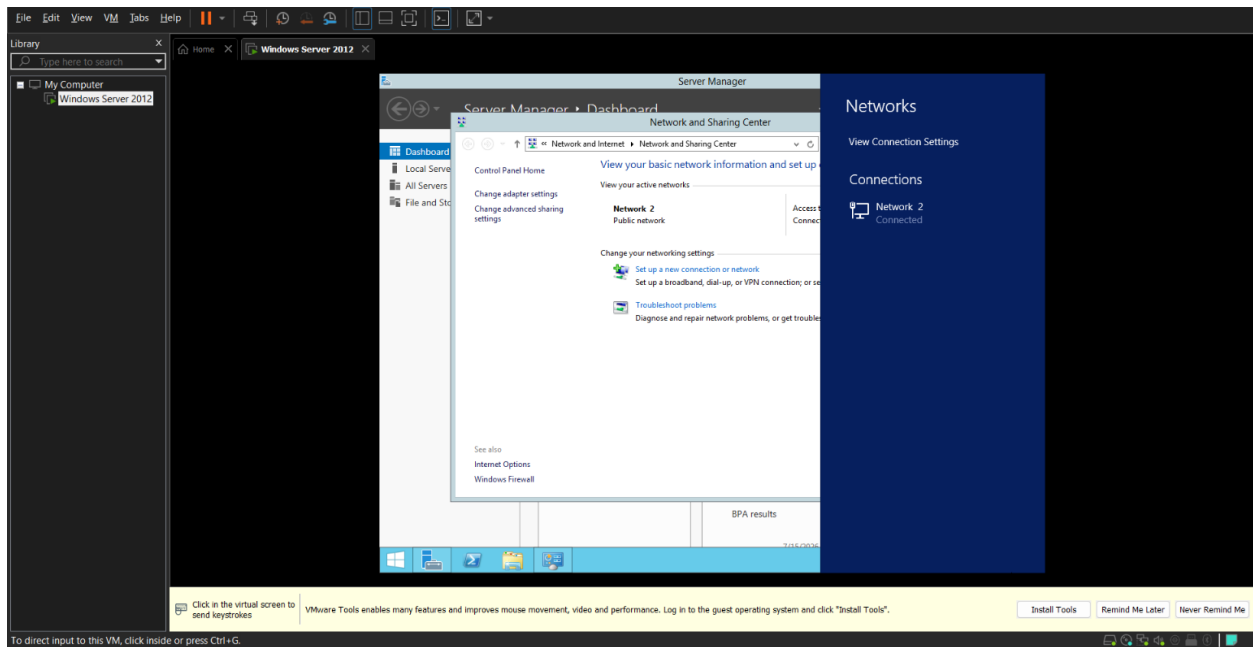
- Now click on **Ethernet** and then click on **Properties**.



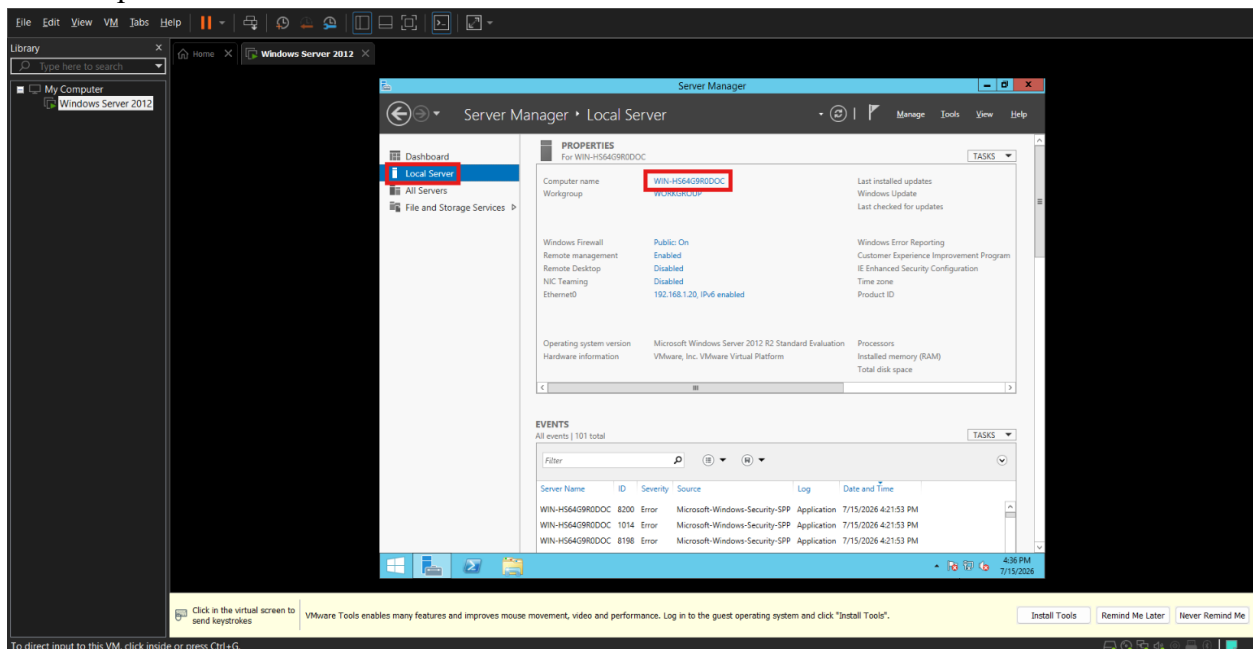
- Now Open **properties of Internet Protocol Version 4 (TCP/IPv4)** and enter the IP Address of your system and other required details as shown and click **OK**.



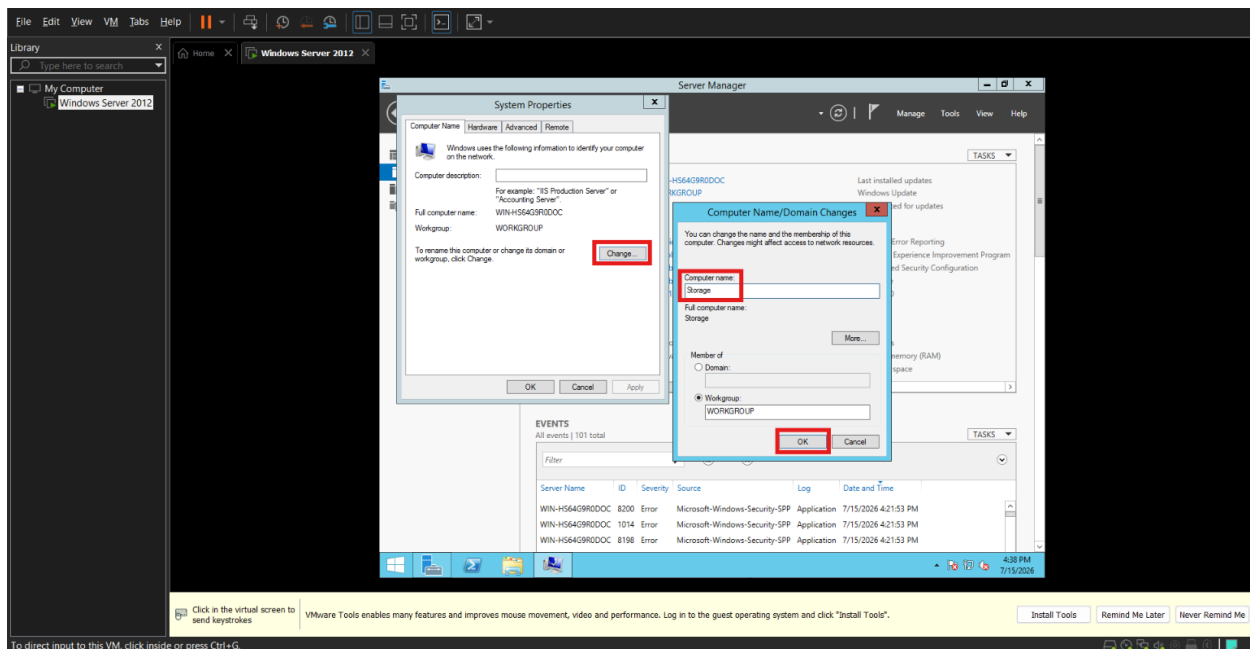
- Click **Yes** for this network prompt and you will be connected to **Internet**.



- Change the name of the computer by opening the Server Manager → Local Server → Computer Name.



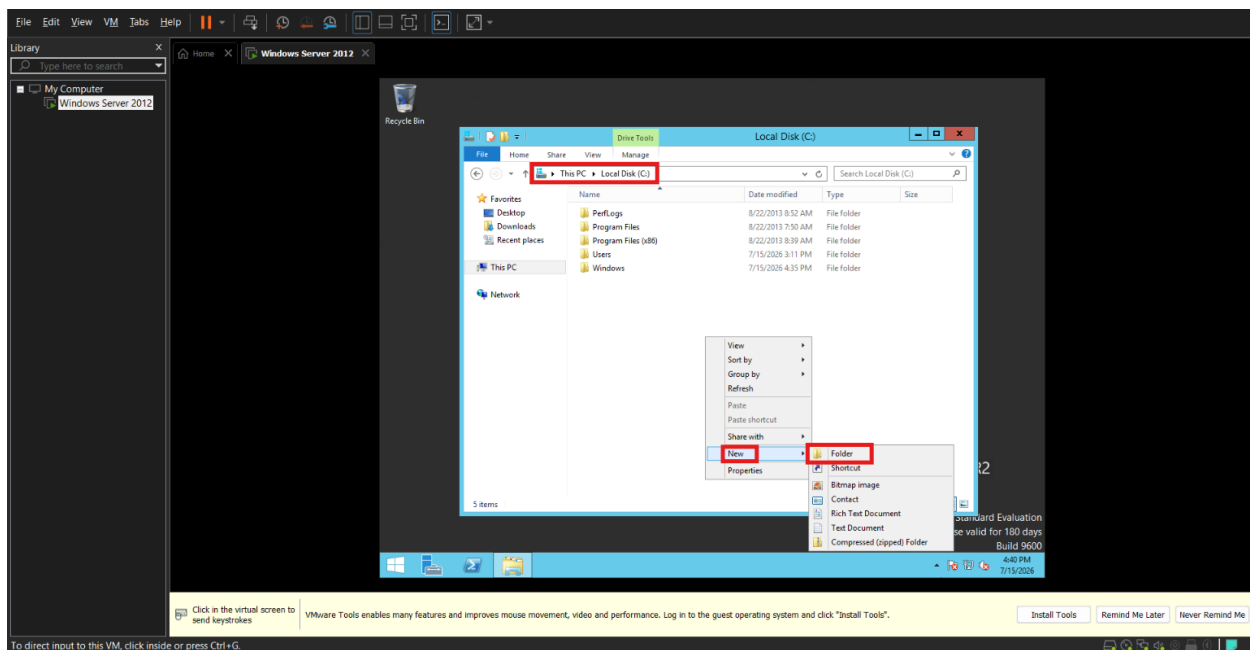
- I will rename it as **Storage** and then **Restart Later**.



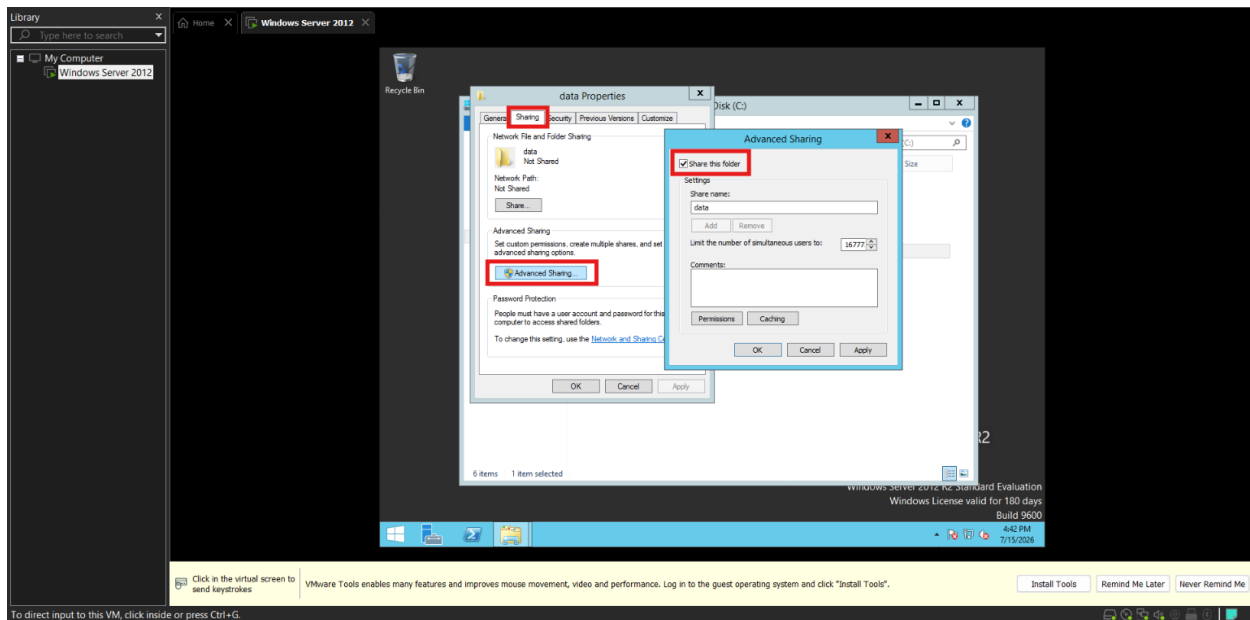
- Create a folder named data.

Step 3 – Create the shared folder

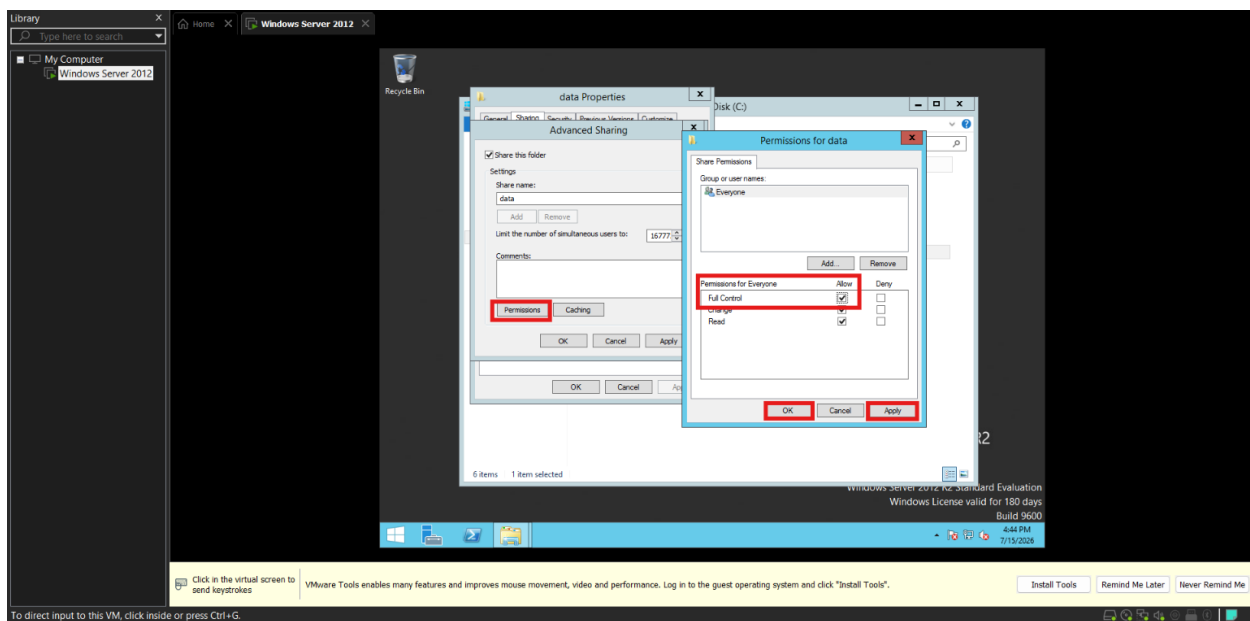
- Open the C: drive.
- Create a folder named **data**.



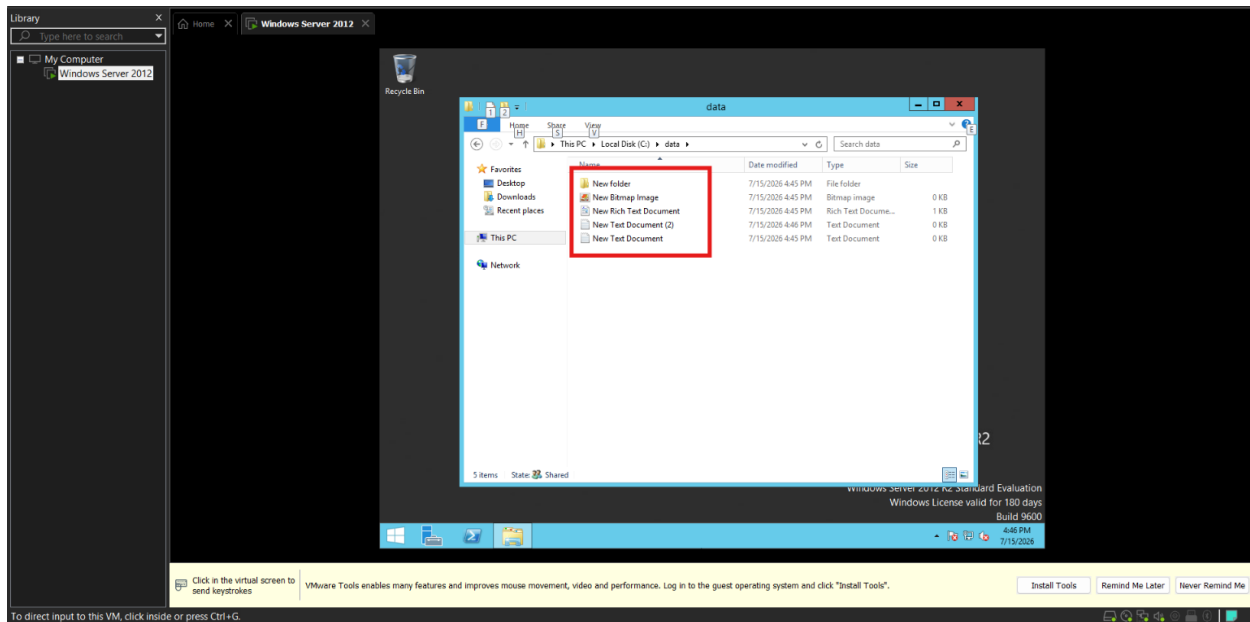
- Open the folder **Properties** by Right-clicking on it and go to sharing tab → Advanced Sharing and click on **Share this Folder**.



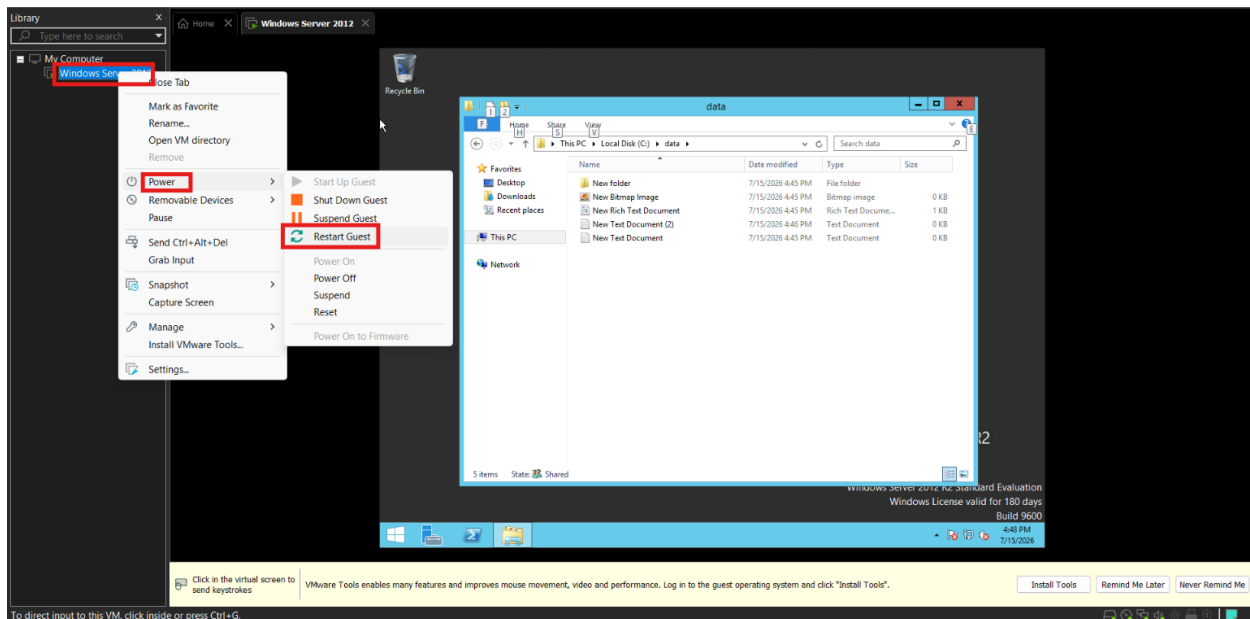
- Click on **Permissions** and give everyone **full control permissions** and click **Apply** and **OK**.



- Add a few sample files into the folder for transfer testing.

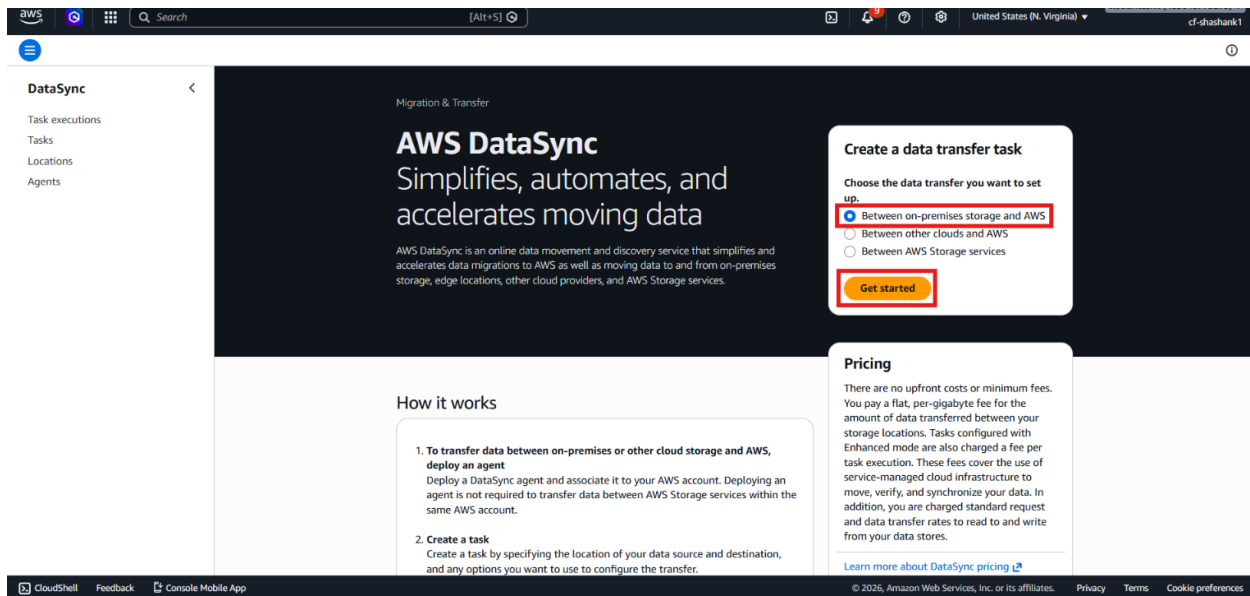


- And then restart the Virtual Machine by Right-clicking on Machine → Power → Restart Guest.

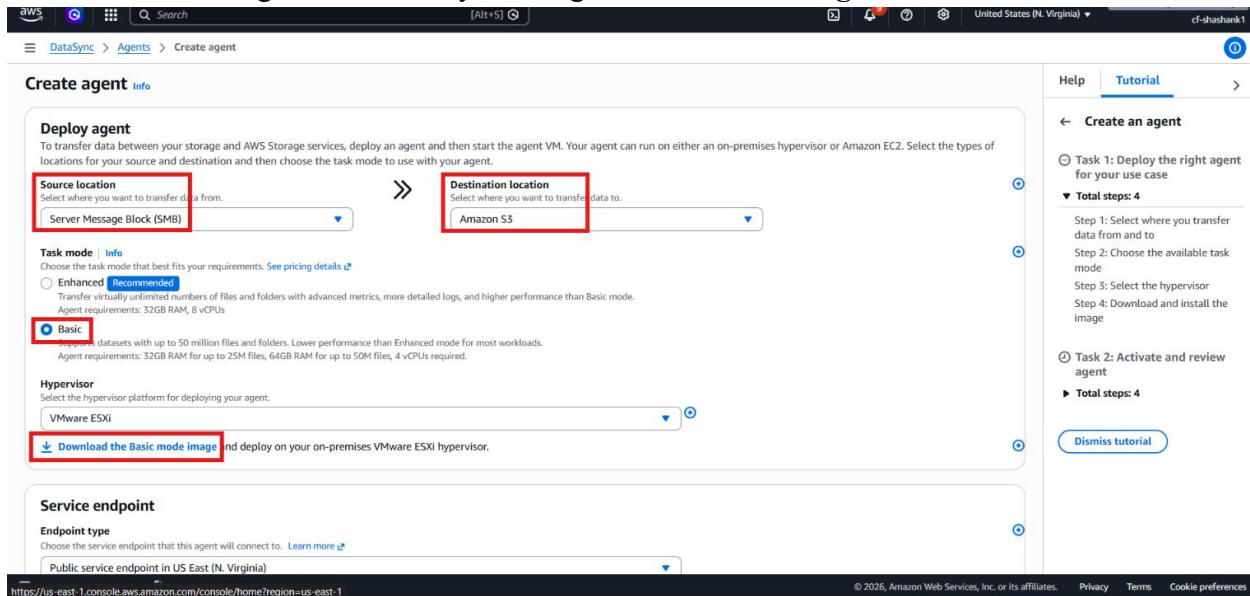


Step 4 – Download and import the DataSync agent

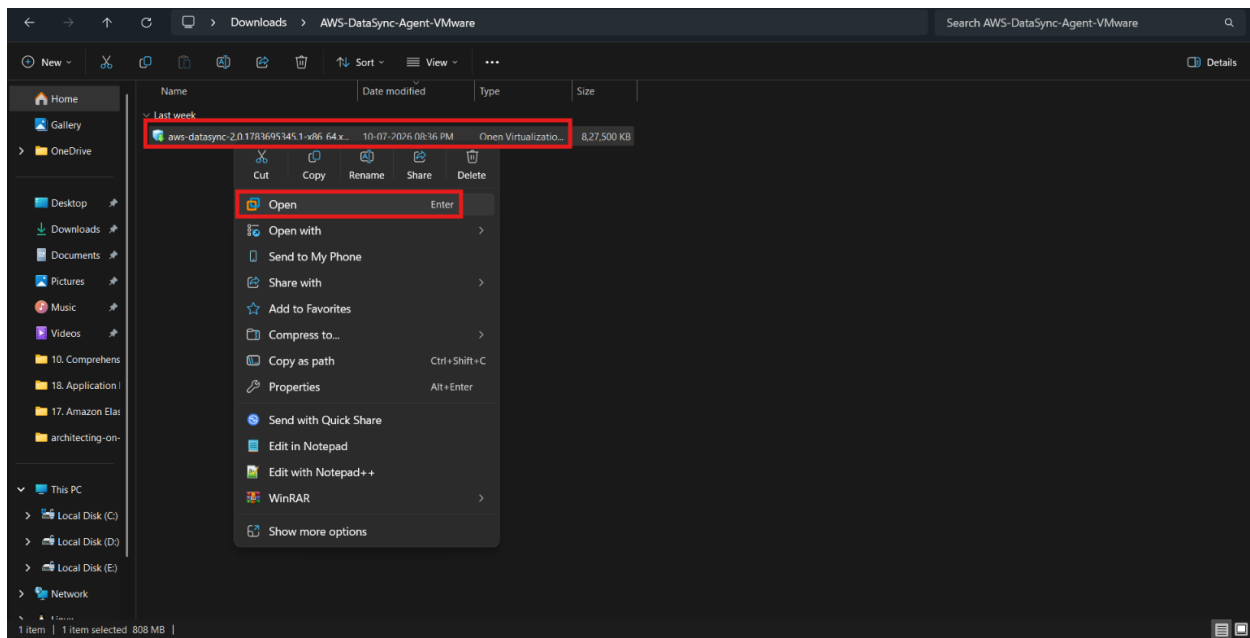
- Open the **AWS Management Console**.
- Navigate to **AWS DataSync**.
- Choose the option for transferring between on-premises storage and AWS.



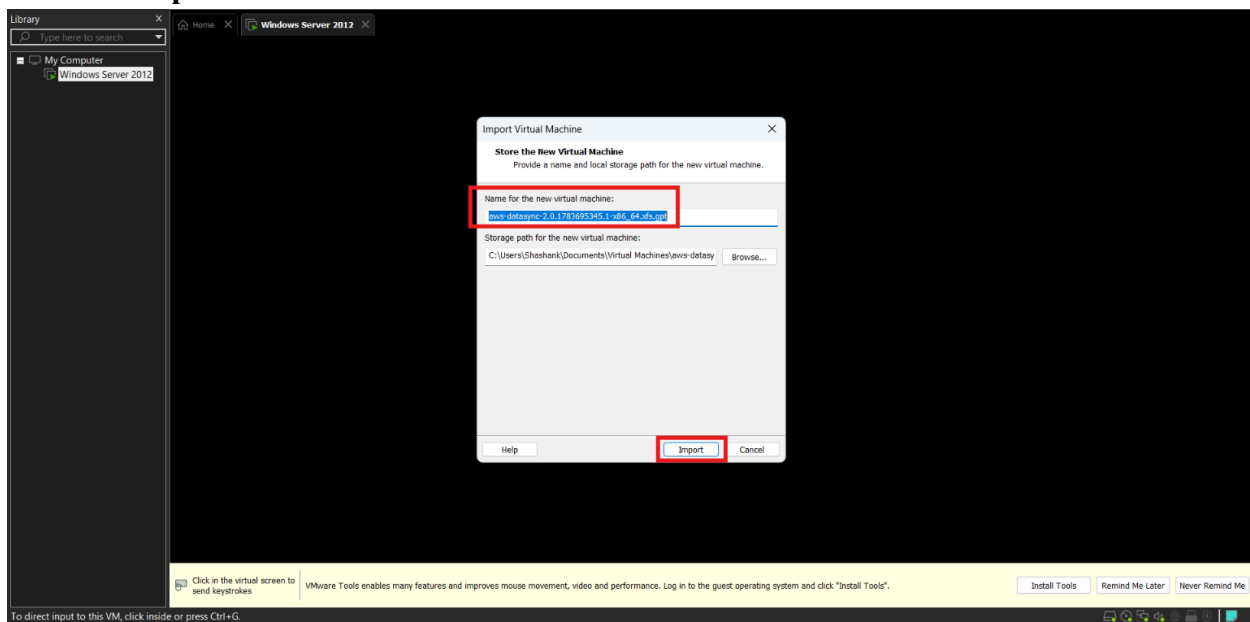
- Select Source and Destination location.
- Select the **VMware ESXi** agent image.
- Download the agent OVA file by clicking on **Download Image**.



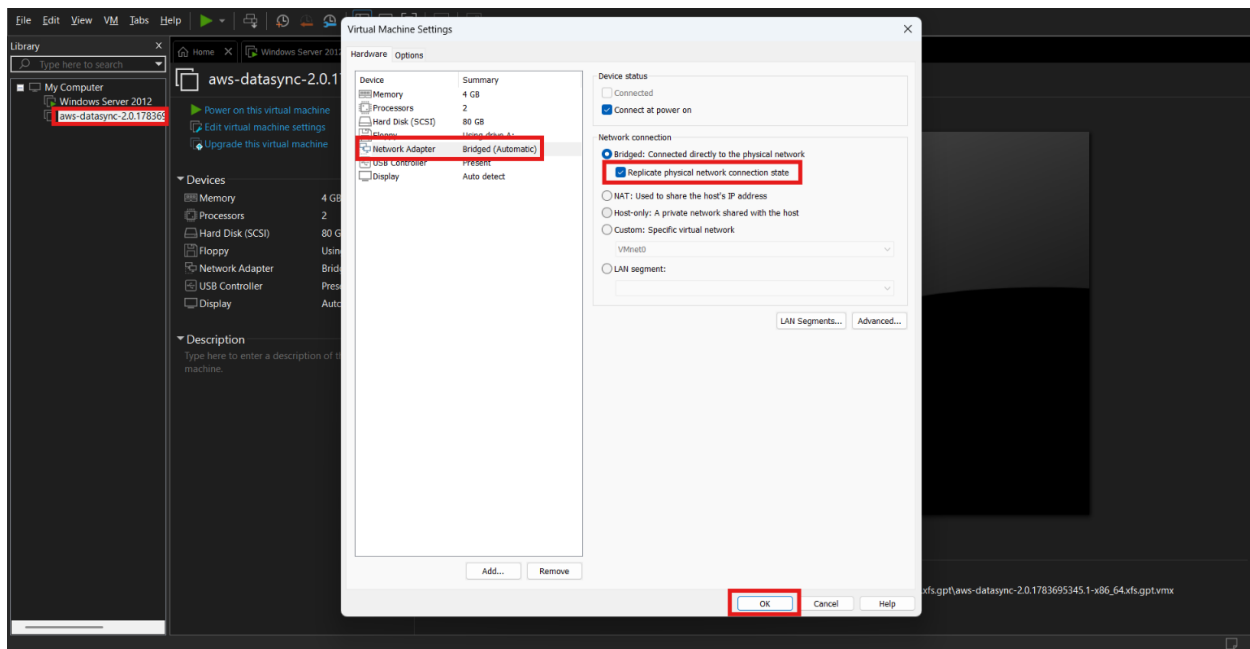
- Extract the zip folder into your computer and then open the extracted folder → **Right-click** on the **OVA** file.



- Click **Import**.



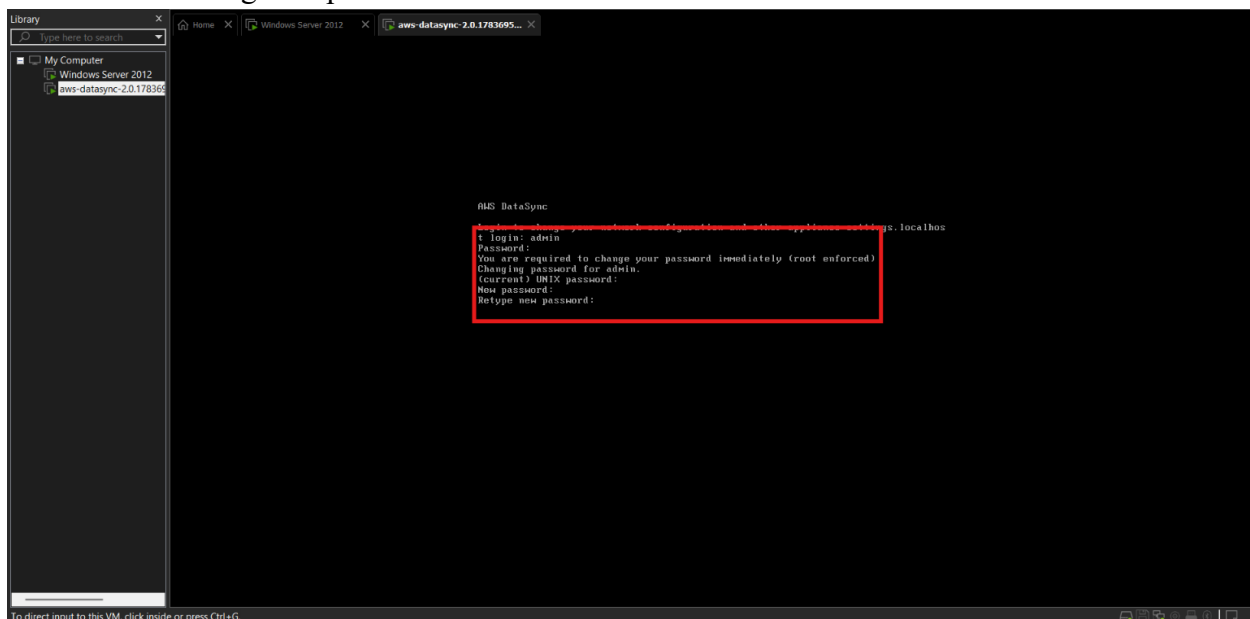
- Enable the same **Replicate physical network connection state** for this also.



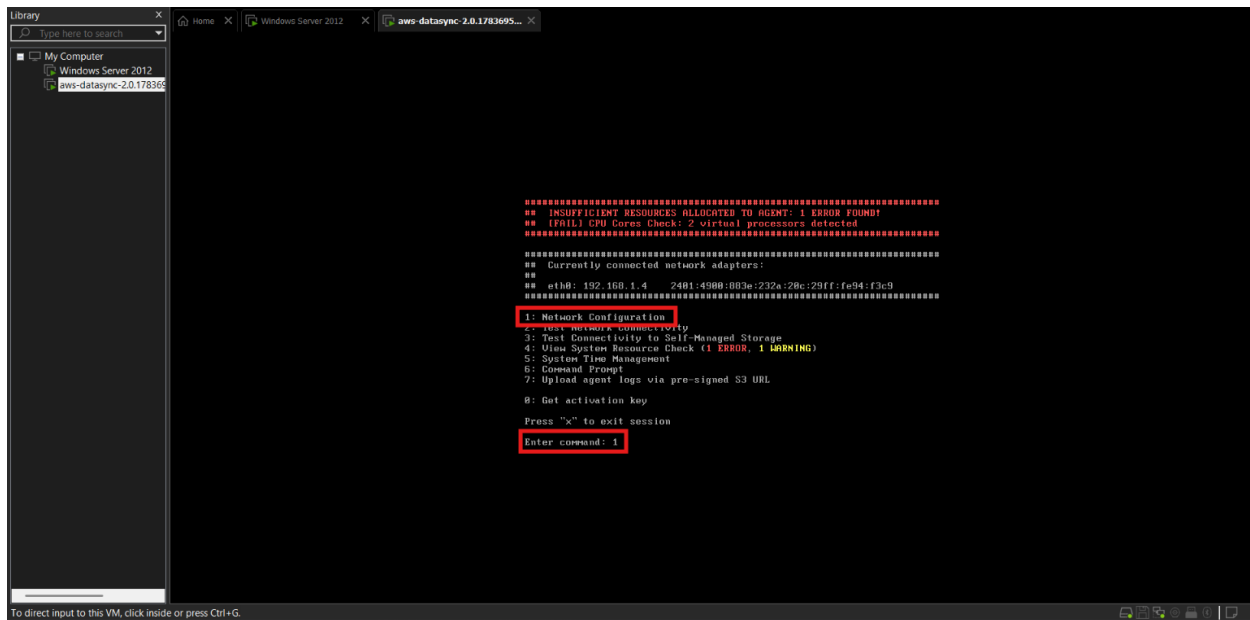
- Start the agent VM.

Step 5 – Configure the DataSync agent network

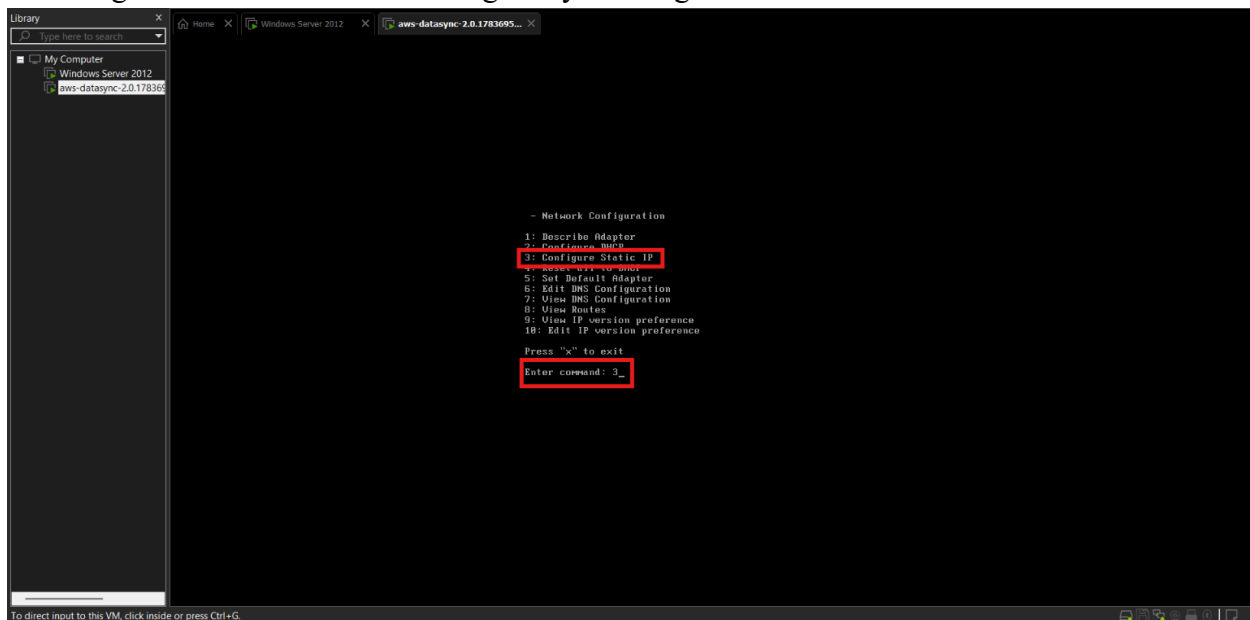
- Log in to the agent with the default admin credentials.
- Username – admin
- Password – password
- And then change the password.



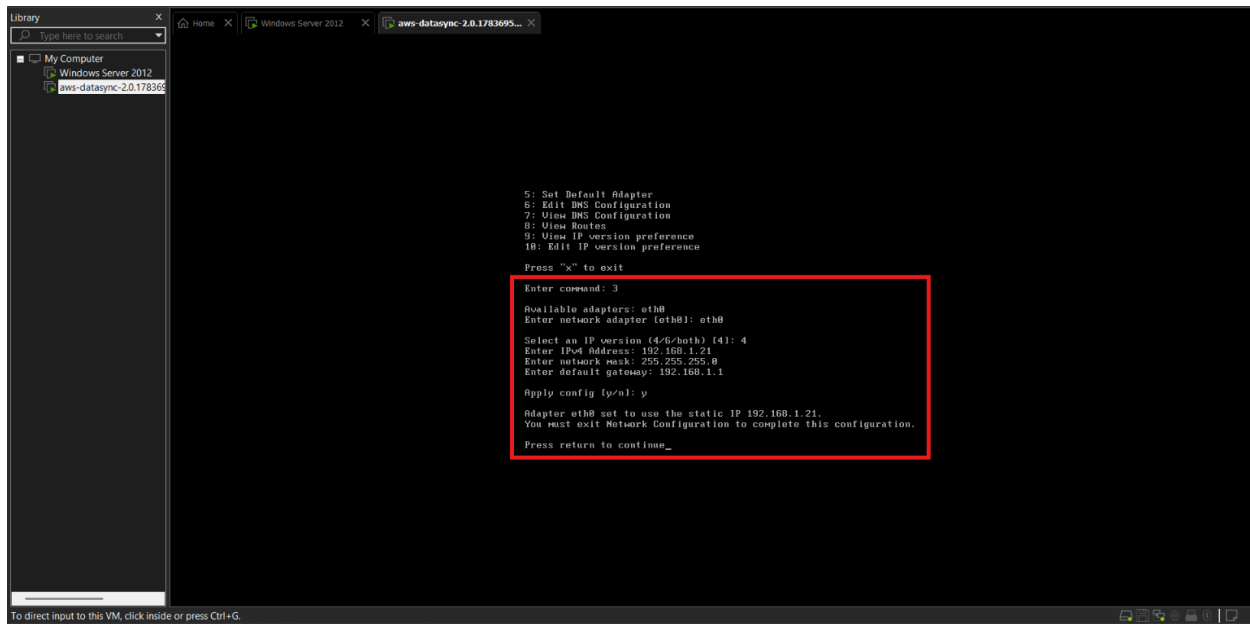
- Open the network configuration menu by entering 1.



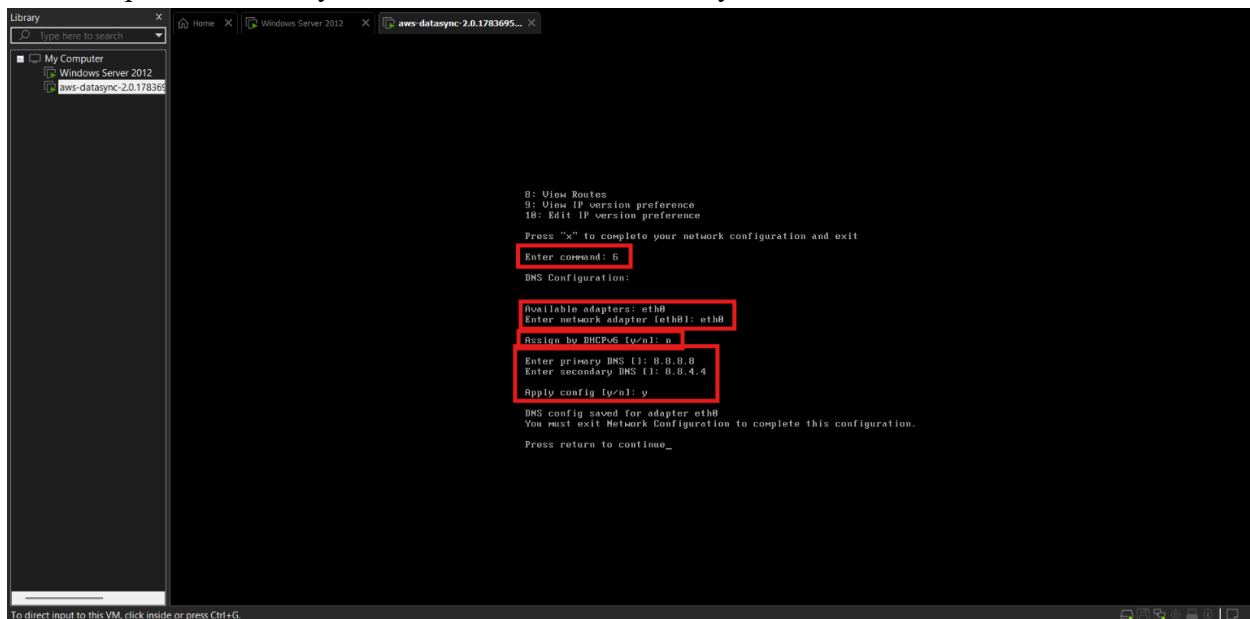
- Assign a static IP address to the agent by entering 3.



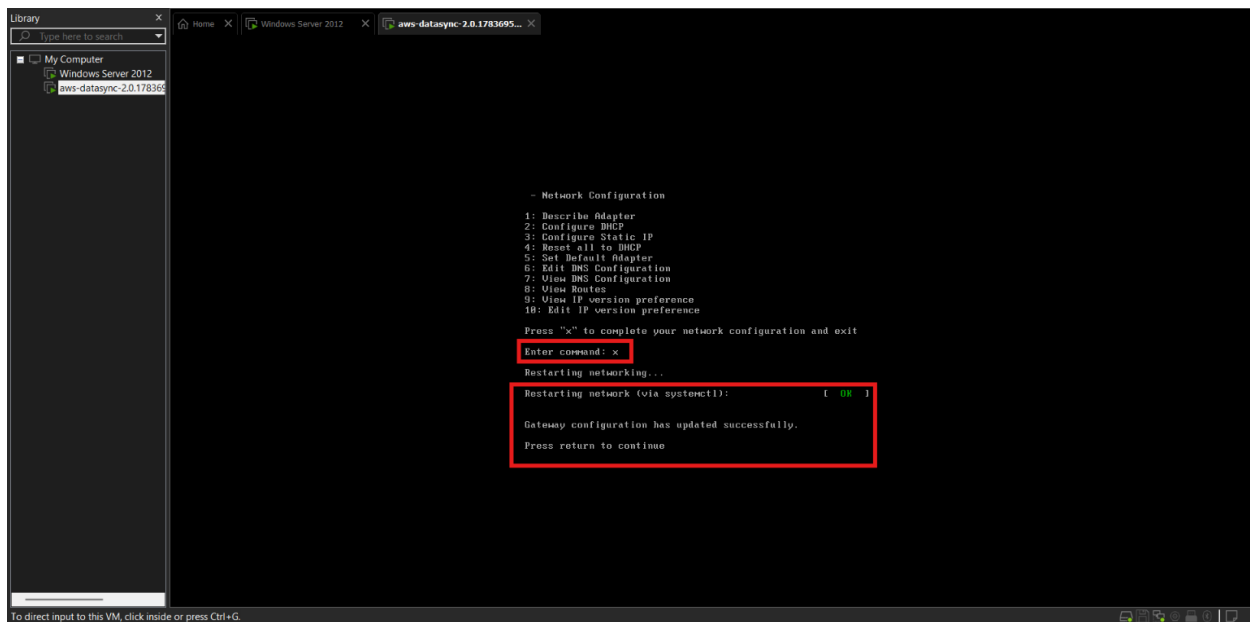
- Use the same set of IP as we used for configuring the Virtual Machine.
- Use the same subnet as the storage server.
- Configure the DNS server settings.



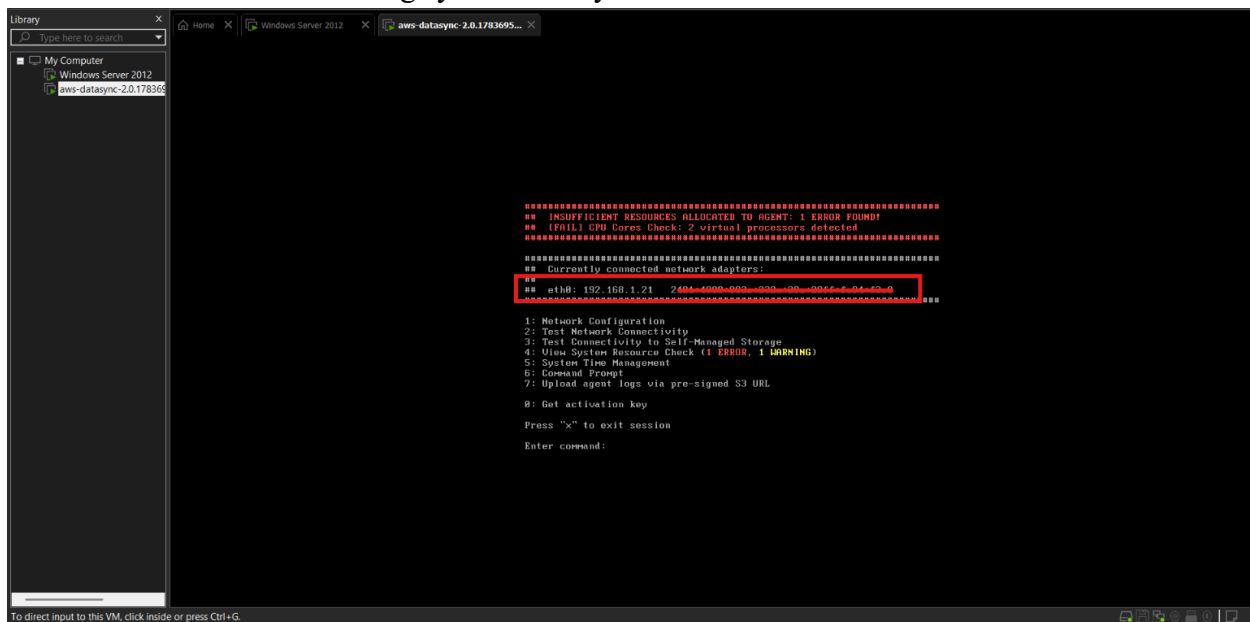
- Similarly, we have to configure the DNS Settings, for that enter 6.
- Now provide Primary DNS as 8.8.8.8 and Secondary DNS as 8.8.4.4



- Save the network configuration by pressing X.



- Here after successful settings you can see your IP Address.



Step 6 – Create the S3 bucket

- Open Amazon S3.
- Create a new bucket named: **amazon-datasync-vm**.

General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)
amazon-datasync-vm

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership
Bucket owner enforced

- Confirm that the bucket is empty before the transfer.

Step 7 – Activate the DataSync agent

- Return to AWS DataSync.
- Choose the **on-premises storage and AWS** option.
- Enter the private IP address of the DataSync agent.
- Click **Get key**.

hypervisor
Select the hypervisor platform for deploying your agent.

VMware ESXi

[Download the Basic mode image](#) and deploy on your on-premises VMware ESXi hypervisor.

Service endpoint

Endpoint type
Choose the service endpoint that this agent will connect to. [Learn more](#)

Public service endpoint in US East (N. Virginia)

Activation key
Activation securely associates the agent that you deployed with your AWS account. [Learn more](#)

Choose a method for obtaining your agent's activation key. [Learn more](#)

☒ **Automatically get the activation key from your agent**
Your browser will connect to your agent over HTTP to get a unique activation key.

☐ **Manually enter your agent's activation key**
Choose this option if you already have an activation key and need to enter it manually.

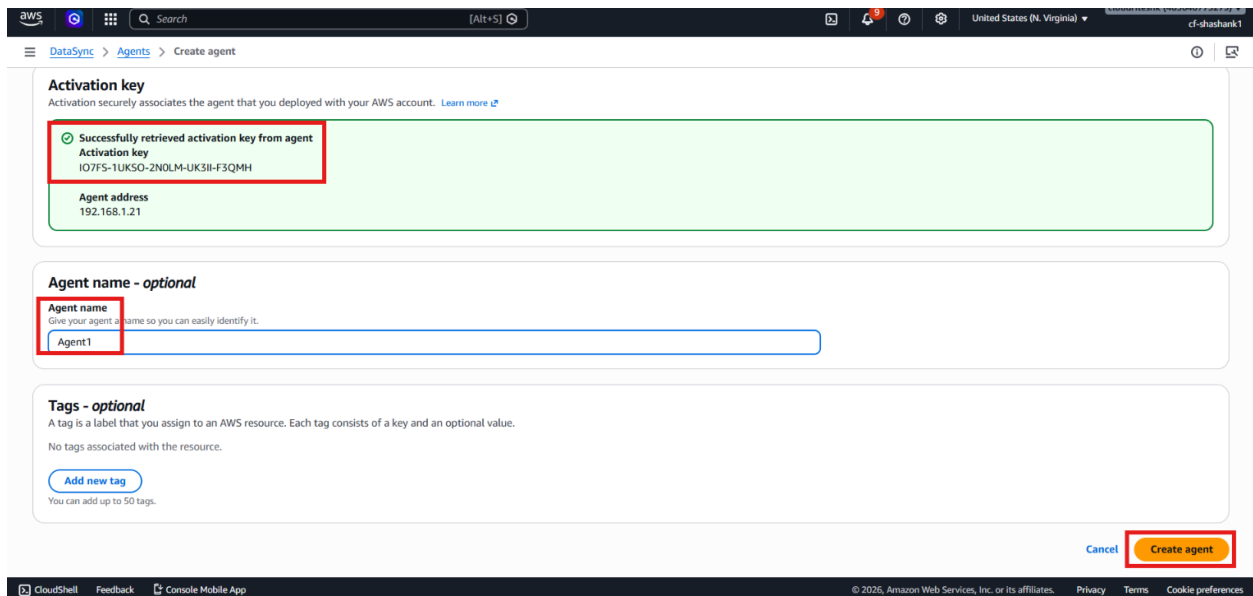
Agent address
When your DataSync agent has been deployed, enter its domain name or IP address. When you click **Get key**, your browser will connect to this address to get a unique activation key.

http:// 192.168.1.21

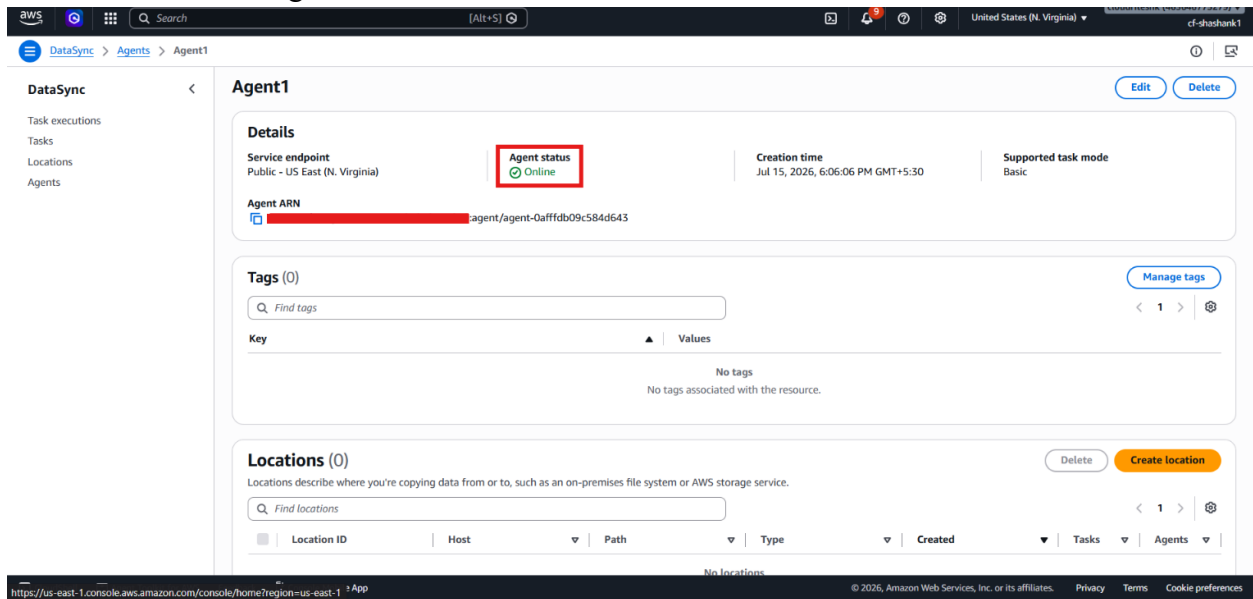
Make sure this address is accessible from your browser.

[Cancel](#) [Get key](#)

- Create and name the agent as **Agent1**.

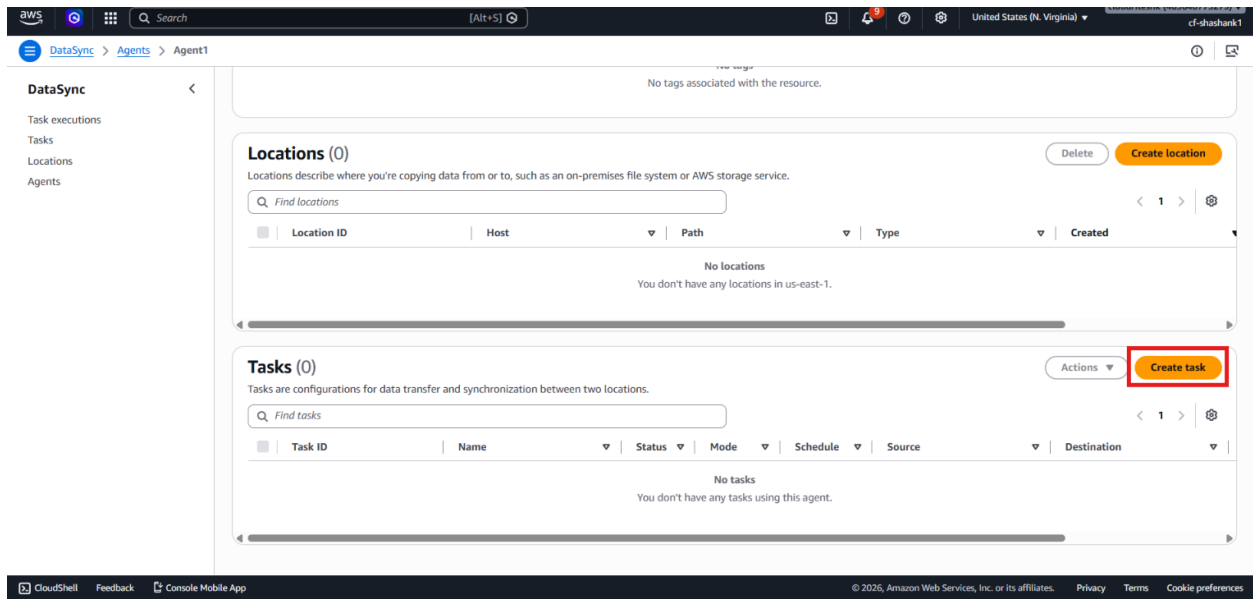


- Confirm that the agent status becomes Online.

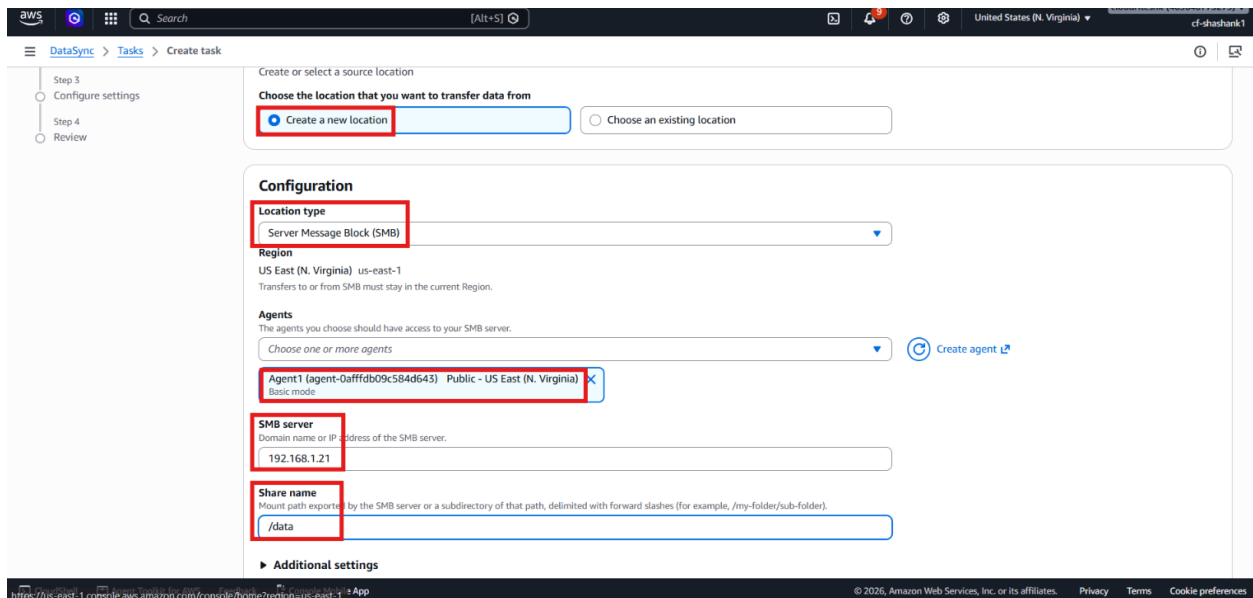


Step 8 – Create the DataSync task

- Scroll down and click **Create task**.



- Choose **SMB** as the source location.
- Enter the SMB server IP address.
- Choose the agent we have just created.
- Enter the share name data.



- Use the administrator username and password we have configured earlier while creating the Virtual Machine.

Authentication info

Authentication type
DataSync authenticates with the SMB server using this method.

☒ NTLM ☐ Kerberos

DataSync authenticates using a user name and a password that you provide. [Learn more about choosing a user name](#) that ensures sufficient permissions to files, folders and metadata.

User
User who has permission to access the SMB server.
administrator

Access configuration
☒ Specify a password ☐ Use a Secrets Manager secret

Password
Password of the user who has permission to access the SMB server.

KMS Key Arn (optional)
Enter an optional KMS Key ARN to encrypt your password.
[Choose an AWS KMS key or enter an ARN](#)

Domain - optional
The name of the Windows domain that the SMB server belongs to.

- Choose the S3 bucket **amazon-datasync-vm** as the destination.
- Let DataSync auto-generate the IAM role by clicking on the **Generate** button.
- Keep the remaining task options unchanged.

Destination location options
Create or select a destination location

Choose the location that you want to transfer data to

☒ Create a new location ☐ Choose an existing location

Configuration

Location type
Amazon S3
Transfers from SMB can go to Amazon EFS, Amazon FSx, or Amazon S3.

Region
US East (N. Virginia) us-east-1
Transfers to or from SMB must stay in the current Region.

S3 URI
s3://amazon-datasync-vm [View](#) [Browse S3](#)

S3 storage class
Choose a storage class you want your objects to be transferred into.
Standard

IAM role
Used to access your S3 bucket. [Learn more](#)
AWSDataSyncS3BucketAccess-amazon-datasync-vm-3d008
View [AWSDataSyncS3BucketAccess-amazon-datasync-vm-3d008](#) on the IAM console.

[Autogenerated](#)

- Enter the name for the Task.
- Choose **Everything** for the Contents to Scan option.
- Set Transfer mode to **Transfer all data**.
- Leave rest settings as default.

Task mode Info

Your task options and performance might vary depending on the task mode you choose. Task mode can't be changed once you create this task. [See pricing details.](#)

☐ Enhanced
Transfer virtually unlimited numbers of objects with advanced metrics, more detailed logs, and higher performance than Basic mode. Currently available for transfers between S3 buckets, between S3 and other clouds when not using an agent, or between NFS or SMB servers and S3 using an Enhanced mode agent.

☒ **Basic**
Transfer objects with standard quotas on the number of files and objects in a dataset. Lower performance than Enhanced mode for most workloads. Requires an agent when copying to or from on-premises or other clouds.

Name - optional

VM-Datasync-Task

Source data options Info

Contents to scan
Specify the data in your source location that you want to transfer to your destination.

Everything

Excludes - optional Info
Specify the files, objects, and folders that you don't want to transfer.

/folder-to-exclude or *.temp-file Add pattern

Transfer options

Transfer mode
Determines whether DataSync transfers only the data and metadata that differ between the source and the destination.

Transfer all data

- Create the task.

Step 9 – Start the transfer

- Open the created task.
- Click **Start** → **Start with defaults**.

Created task

VM-Datasync-Task

Start **Stop** **Edit** **Delete**

Start with defaults
Start with overriding options

Task status **Available**

Latest execution -

Task mode Basic

Task ARN `arn:aws:datasync:us-east-1:123456789012:task/task-03ce64561bb1862ab`

Creation time Jul 16, 2026, 12:00:24 PM GMT+5:30

Locations **Options** **Schedule** **Tags** **Monitoring** **Task executions**

Source location

Location ID `loc-0f30d324564b5b921`

Region US East (N. Virginia) us-east-1

Type Server Message Block (SMB)

Path /data/

Server 192.168.1.21

Agents `agent-0afffb09c584d643`

Authentication type NTLM

User admin

Secret `aws-datasyncloc-0f30d324564b5b921-eNoRjw`

SMB version Automatic

Destination location

Location ID `loc-0a07990cade1d16d9`

Region US East (N. Virginia) us-east-1

Type Amazon S3

Path /

Creation Time Jul 16, 2026, 12:00:23 PM GMT+5:30

S3 bucket `amazon-datasync-vm`

S3 storage class Standard

S3 bucket IAM role `AWSDataSyncS3BucketAccess-amazon-datasync-vm-3d008`

- Keep the default execution settings.
- Wait for the transfer to complete.

Step 10 – Verify the copied files

- Open the task history.
- Confirm that the transfer completed successfully.

- Check how many files were copied.

The screenshot shows the AWS DataSync console for a task named 'exec-0034a47dc37abfa80'. The 'Overview' section indicates the 'Execution status' is 'Success'. The 'Performance' section shows that 6 files were transferred, with a data throughput of 0.36 B/s and a file throughput of 0.31 file/s. The task execution ARN is 'arn:aws:datasync:us-east-1:123456789012:task/task-05d020729b0b5b593/execution/exec-0034a47dc37abfa80'.

- Open the S3 bucket.
- Refresh the bucket contents.
- Confirm that the files from the shared folder now appear in Amazon S3.

The screenshot shows the Amazon S3 console for the bucket 'amazon-datasync-vm'. The 'Objects' tab is selected, showing a list of 6 objects. The objects are: '.aws-datasync-metadata' (aws-datasync-metadata), 'New Bitmap Image.bmp' (bmp), 'New folder/' (Folder), 'New Rich Text Document.rtf' (rtf), 'New Text Document (2).txt' (txt), and 'New Text Document.txt' (txt). The objects are listed with their names, types, last modified dates, sizes, and storage classes.

Verification

- Verified that the Windows Server virtual machine was created successfully in **VMware Workstation**.
- Verified that the shared folder data was created and populated with sample files.
- Verified that the **AWS DataSync** agent was downloaded, imported, and activated successfully.
- Verified that the SMB source location pointed to the correct storage server and share.

- Verified that the destination **Amazon S3** bucket on prem data was created successfully.
- Verified that the DataSync task completed successfully.
- Verified that the files were copied from the SMB share into **Amazon S3**.